



ECOLE
POLYTECHNIQUE
DE BRUXELLES

02. Mathematical Foundations I

INFO-H502

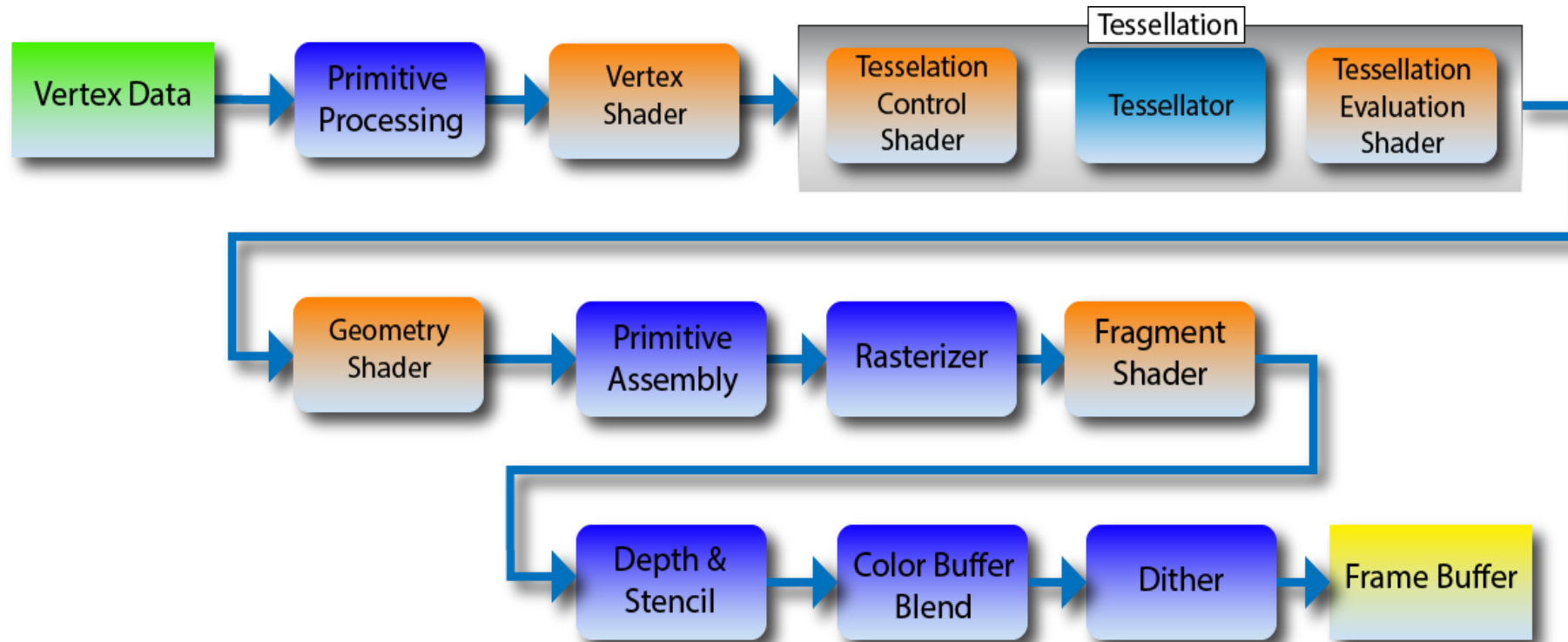
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LISA-VR

2024

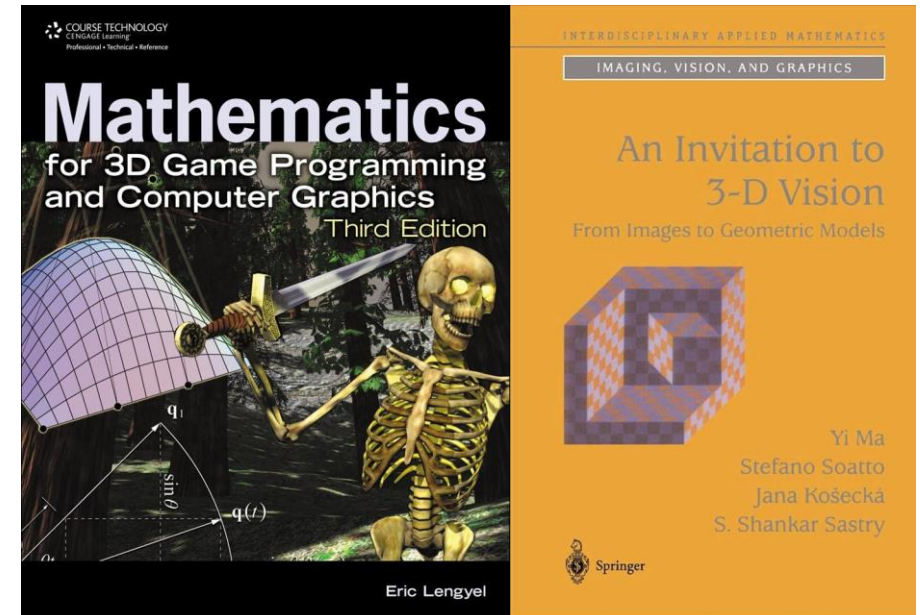
Previously

- The OpenGL Pipeline to render 3D objects to the 2D screen



Today

- More mathematical lesson
 - I will often write **why?** Near an equation, you should work on your own to understand !
- Most of the maths you will ever need in computer graphics
 - Linear algebra
 - Geometry
- The presentation in the slides is self-contained, but it is mainly a summary of:
 - ISA (Geometry chapters)
 - Mechanics (1 and 2)
 - Linear Algebra
- Inspired from:
 - mathematics for 3D game programming and computer graphics
 - An invitation to 3-D Vision

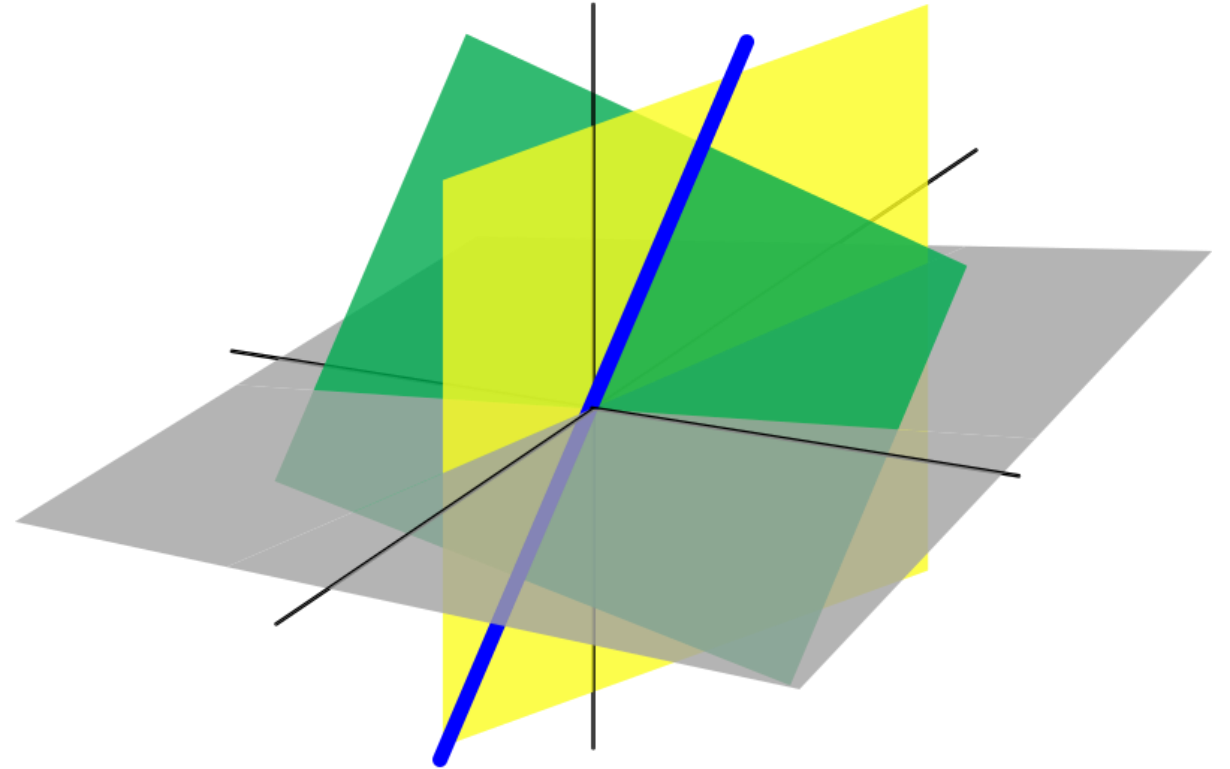
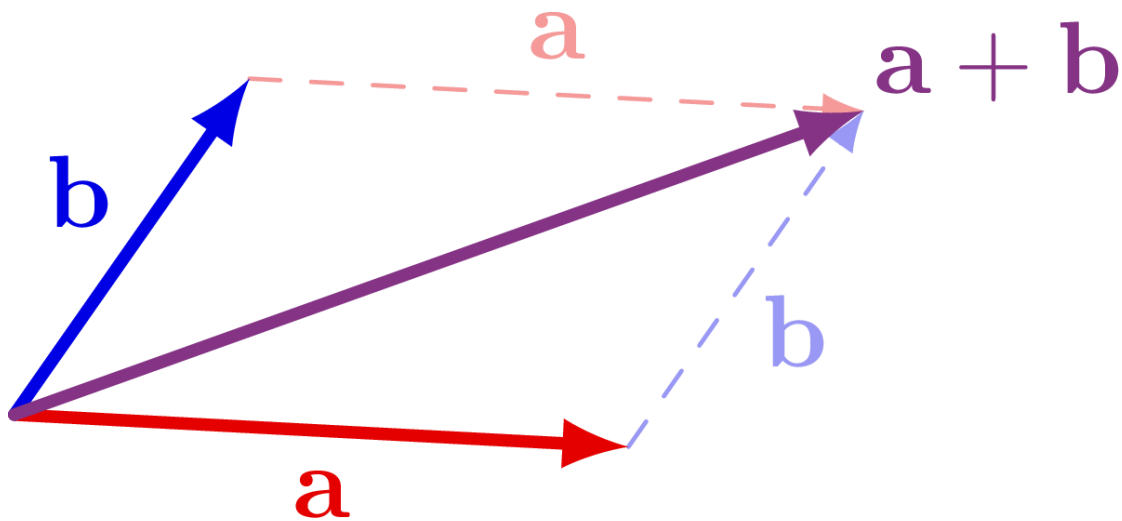


What I Expect for the exam

- After this lesson I expect that:
 - You know the main Linear Algebra structures that we use
 - You have a good command of the mathematics contained in those slides
 - From a theoretical point of view, being able to do the main proofs
 - You are able to solve exercises that apply this theory
 - The math are mostly first year ISA and mechanics, so if you do not feel comfortable with the maths, please, review the exercises sessions in ISA
 - You are able to reproduce the complete « summary » of how everything fits together
- This and the next lessons are the most mathematical lessons, there will be maths in other sessions, but those are the most important ones.

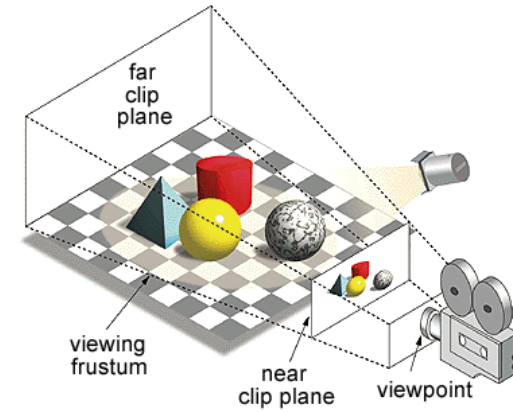
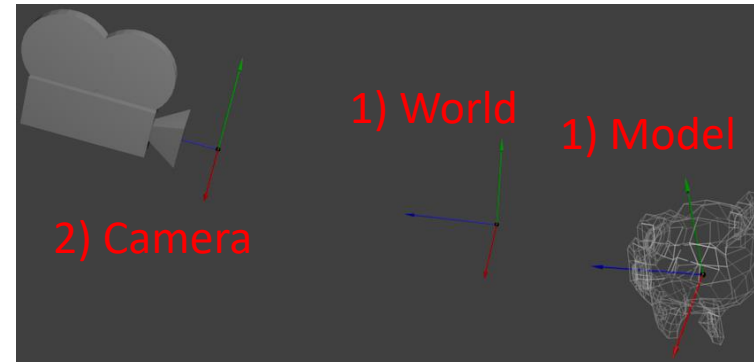
Why Linear Algebra ?

- Bridge between **geometry**, **physics** and **computations**
- In computer graphics, most things can be expressed by matrices
- Computers are extremely fast with matrix operations (numerical algebra)
- We will transform all our operations into matrices !

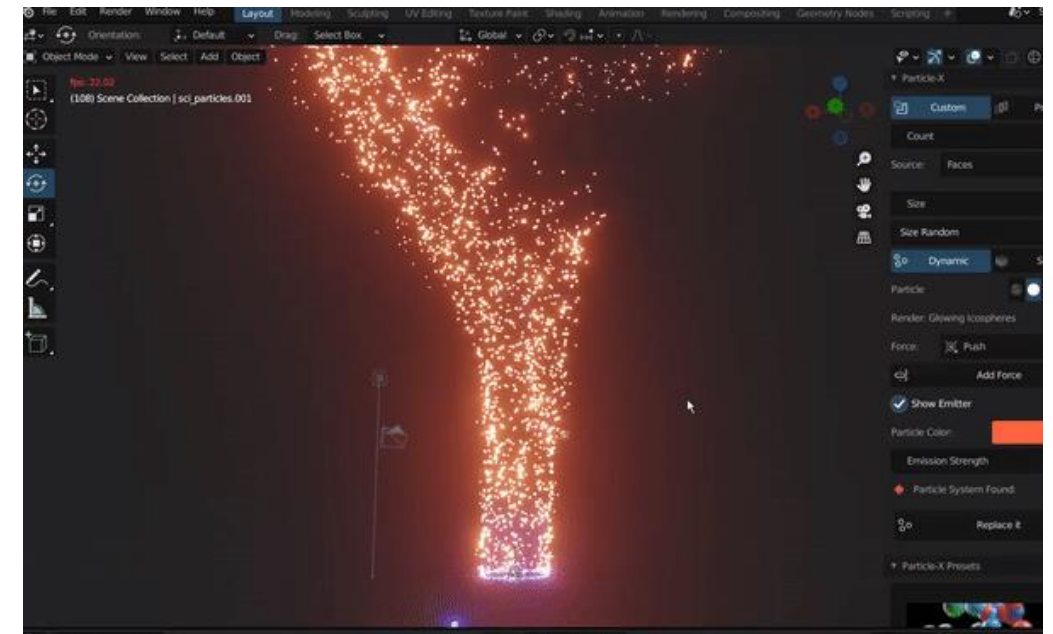


Where is Linear Algebra applied ?

- To move meshes' vertices in space
 - Rotations and translations
- To do perspective renderings
- To simulate physics
- To compute ray interactions for raytracing
- Etc.

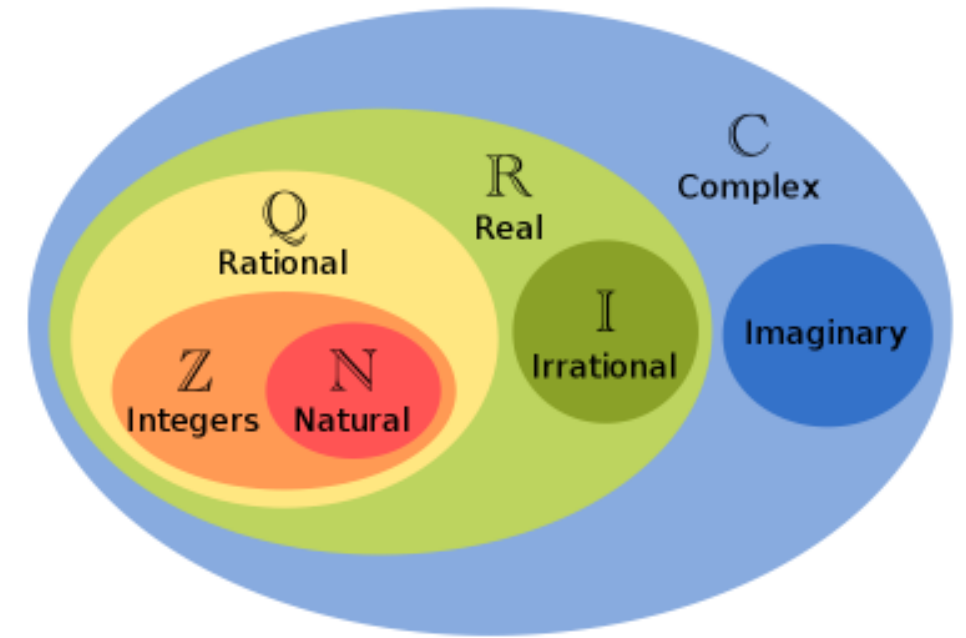


- To do that, let's first dive into Algebra !
- We will start slow with the basics:
 - What is a point in space exactly?

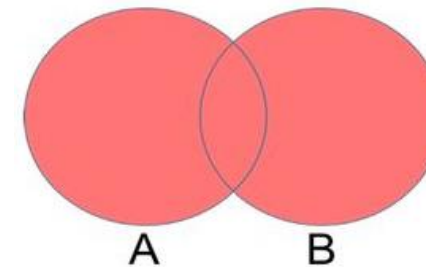
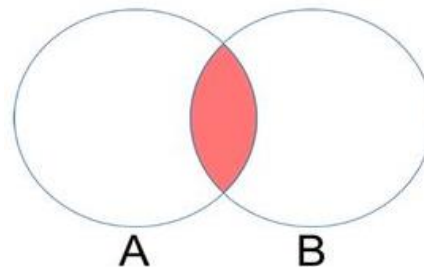
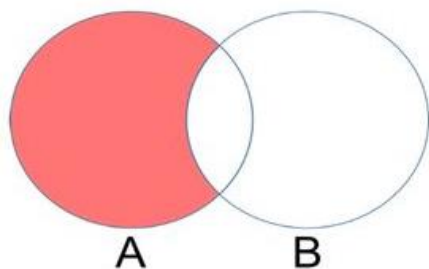


Sets – Our first object

- *Definition*: A **set** is a collection of objects.
(Informally)
- You should be used to sets such as
 - $\emptyset, \{1,2\}, \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, etc..
- Set do not have operations on their objects
- They are just collections of objects.
- We still can create novel subsets using set operations:
 - $A \cup B, A \cap B, A \setminus B, A^c$

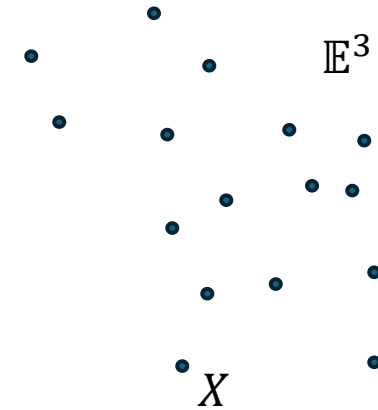


Which operation is which image ?



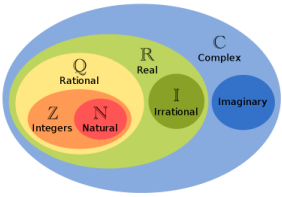
Euclidean Space $\mathbb{E}^3$

- The **Euclidean space**, is a *set* of 3D points in space (without coordinates)
- Every point $X \in \mathbb{E}^3$ is represented as
 - $X = (x_1, x_2, x_3)^T \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}$
- Not very useful as there is only a topology*
- We need operations to measure distances and angles
- Let's build all the structures needed to do that !



* a set of neighbourhoods for each point that satisfy some axioms formalizing the concept of closeness

Summary

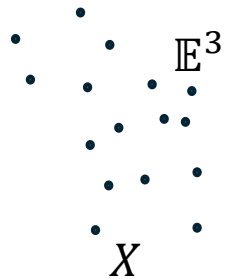


Sets



Eléments = points

Euclidean Space $\mathbb{E}^3$
A set of points



We have only topology

- For the moment, we have a set of points in space, with only topology

Groups – Our first structure

- Groups were created for handling, in a unified way, many mathematical structures such as:
 - Numbers $(\mathbb{Z}, +)$, $(\mathbb{R}, .)$, $(\mathbb{Q}, .)$
 - Geometric shapes $(D_n, \circ)$
 - Polynomial roots,
 - etc.
- Several groups are relevant for computer graphics.
 - They encapsulate the properties of some mathematical object under a unique name, allowing for **easier communication**, and debugging.

Check that they form groups !



D_n is the Dihedral group of order n ,
i.e, the set of symmetries of a regular n -sided polygon, including rotations and reflections

Groups

Mnemo: CANI

- **Definition:** A **group** is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \rightarrow G$ is a map
 - **Associativity** $\forall a, b, c \in G : (a \cdot b) \cdot c = a \cdot (b \cdot c)$
 - **Neutral element** $\exists e \in G \mid \forall a \in G : e \cdot a = a \cdot e = a$
 - **Inverse element** $\forall a \in G, \exists b \in G \mid a \cdot b = b \cdot a = e$
- The group is called abelian if:
 - **Commutativity** $\forall a, b \in G : a \cdot b = b \cdot a$
- You are used to $\mathbb{N}, \mathbb{Z} = \{-10, \dots, -2, -1, 0, 1, 2, \dots\}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
- We will learn $SE(3), SO(3), GL(3), O(3), SP(3), U(3), SU(3), \mathbb{Q}_8, \dots$

Name	Symbol	Lie algebra
Euclidean space	i^n	i^n
General linear group	$GL(n, F)$	$\mathfrak{gl}(n, F)$
Special linear group	$SL(n, F)$	$\mathfrak{sl}(n, F)$
Orthogonal group	$O(n, F)$	$\mathfrak{o}(n, F)$
Special orthogonal group	$SO(n, F)$	$\mathfrak{so}(n, F)$
Symplectic group	$SP(n, F)$	$\mathfrak{sp}(n, F)$
Unitary group	$U(n)$	$\mathfrak{u}(n)$
Special Unitary group	$SU(n)$	$\mathfrak{su}(n)$
(p, q) type Unitary group	$U(p, q)$	$\mathfrak{u}(p, q)$
Special (p, q) type Unitary	$SU(p, q)$	$\mathfrak{su}(p, q)$

Fields

Mnemo: CANI CANI D

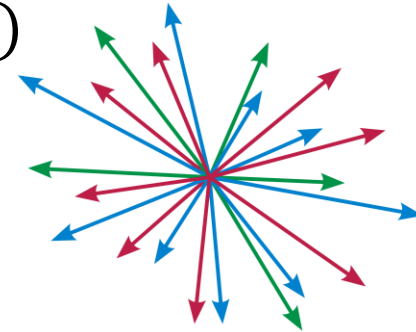
- **Definition:** A **field** is a triple $(K, +, \cdot)$ where K is a set and $+, \cdot$ are maps $K \times K \rightarrow K$ satisfying:
 - $(K, +)$ is an abelian group (CANI)
 - $(K \setminus \{0\}, \cdot)$ is an abelian group (CANI)
 - The maps $+$ and $\cdot$ satisfy the distributive property:
 - **Distributive property** $\forall a, b, c \in K: (a + b) \cdot c = a \cdot c + b \cdot c$
- Examples: $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$ Check that they form fields !
- They allow to work with the common addition, subtraction, multiplication and division operations over a set.

Note: in French a Field is called a Corps,
and Corps Gauche if there is no commutativity for $(K \setminus \{0\}, \cdot)$

Vector spaces over a field K

Mnemo: CANI ADDU

- **Definition:** Let $(K, +, \cdot)$ be a field. A **vector space over K** is a triplet $(V, \oplus, \odot)$ where V is a set and $\oplus: V \times V \rightarrow V$ and $\odot: K \times V \rightarrow V$ are maps satisfying the following axioms:
 - $(V, \oplus)$ is an abelian group (CANI)
 - **Associativity** $\forall a, b \in K: \forall v \in V: (a \cdot b) \odot v = a \odot (b \odot v)$
 - $\odot$ **Distribute** $+$ $\forall a, b \in K: \forall v \in V: (a + b) \odot v = (a \odot v) + (b \odot v)$
 - $\odot$ **Distribute** $\oplus$ $\forall a \in K: \forall v, w \in V: a \odot (v \oplus w) = (a \odot v) \oplus (a \odot w)$
 - **Unitary** $\forall v \in V: 1 \odot v = v$
- Vector spaces are also called linear spaces
- The elements of V are called vectors, the elements of K are called scalars
- The map $\odot: K \times V$ is called scalar multiplication
- Example: The solutions set of $Ax = 0$ Why?

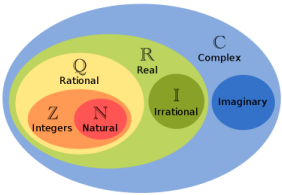


Summary

Add a law
 $+: G \times G \rightarrow G$

Add a law $.: G \times G \rightarrow G$
 And $.$ distribute $+$.

Objects = vectors
 Add 2 laws + inherit field ops
 $\oplus: V \times V \rightarrow V$
 $\odot: K \times V \rightarrow V$

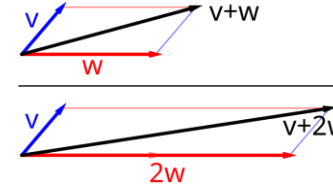


+	0	1	2	3
0	0	1	2	3
1	1	0	3	2
2	2	3	0	1
3	3	2	1	0

Addition Table

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	4	6
3	0	3	6	5

Multiplication Table



Sets

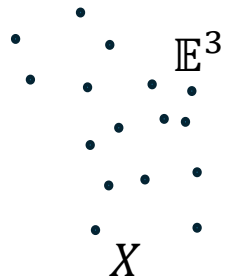
Groups $(G, .)$

Field $(K, +, .)$

Vector space $(V, \oplus, \odot)$

Eléments = points

Euclidean Space $\mathbb{E}^3$
 A set of points

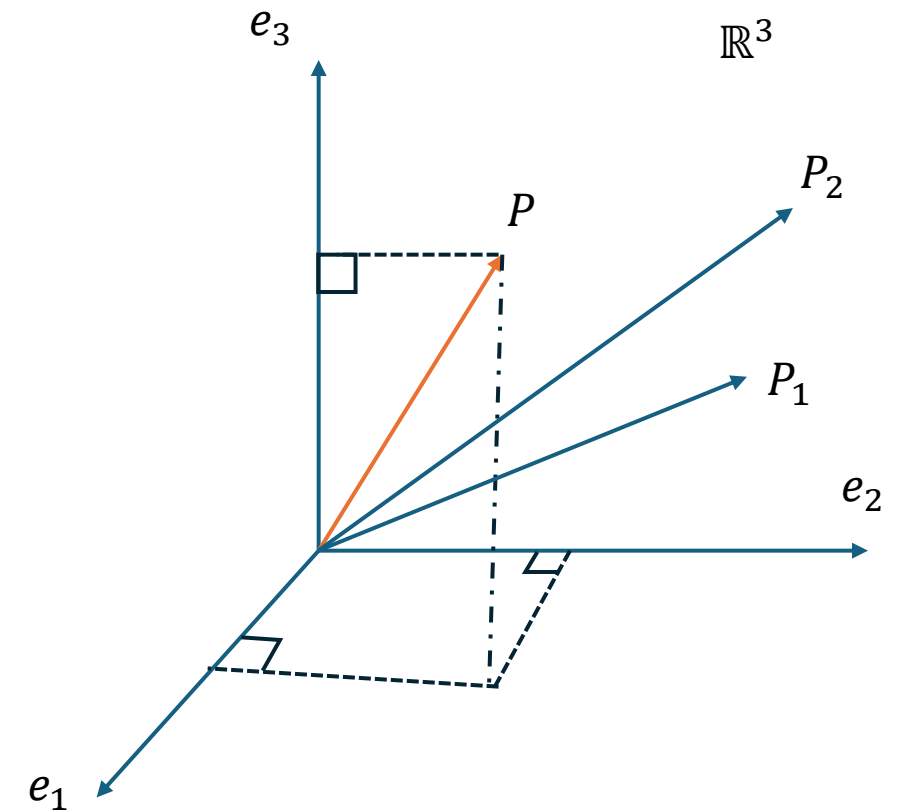


We have only topology

- We can now build the 3D space as we know it

Three Dimensional Space ($\mathbb{R}^3, +, \cdot$)

- 3D space has an orthonormal basis system:
 - $e_1 = (1, 0, 0)^T, e_2 = (0, 1, 0)^T, e_3 = (0, 0, 1)^T$
- A point P in $\mathbb{R}^3$ is defined as
 - $P = (x, y, z)^T \in \mathbb{R}^3 = xe_1 + ye_2 + ze_3$
It is also a vector from the origin !



We should remember that points and vectors do not always live in the same space

Three Dimensional Space $\mathbb{R}^3$ – Metric

- To measure projections and angles, we endow $\mathbb{R}^3$ with a **dot product** $\cdot : V \times V \rightarrow K$:

$$\bullet e_i \cdot e_j = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

- Inner product (= dot product = a projection):

$$\bullet v \cdot w = v_1 w_1 + v_2 w_2 + v_3 w_3 \in \mathbb{R} = |v| |w| \cos \alpha \quad \forall v, w \in \mathbb{R}^3$$

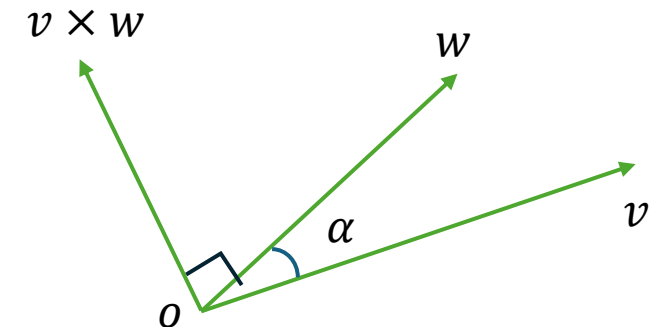
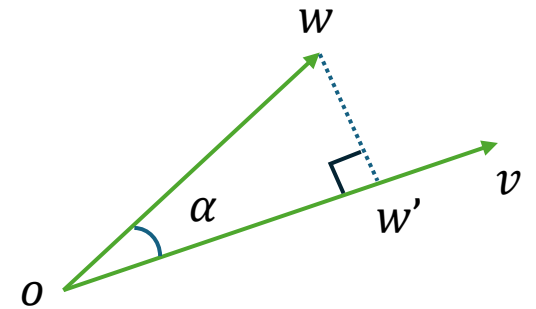
Why ?

- **Cross-product:**

Note: $|v|$ is not yet defined !

$$\bullet v \times w = \begin{vmatrix} e_1 & e_2 & e_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}, \text{ a perpendicular vector} \quad \forall v, w \in \mathbb{R}^3$$

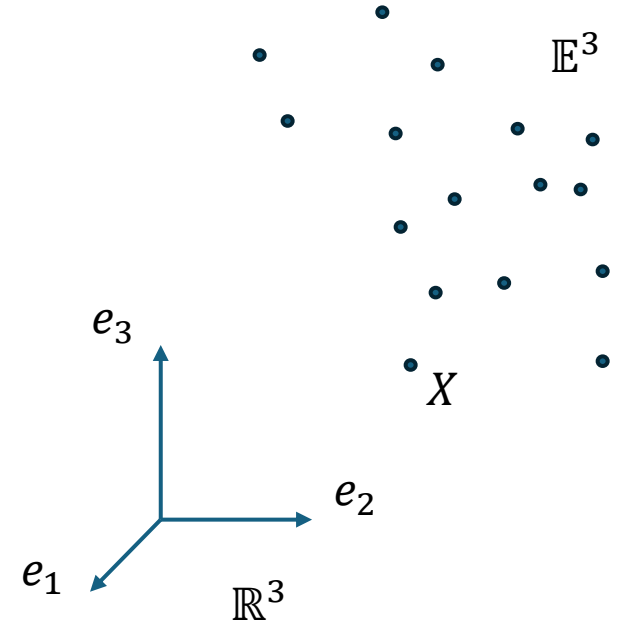
- The length of this vector corresponds to the area of the parallelepiped generated by the 3 axis.
- $|v \times w| = |v| |w| |\sin \alpha|$



From Euclidean Space $\mathbb{E}^3$ to vector space $\mathbb{R}^3$

- The Euclidean space, is a 3D space of points (without coordinates)
- Every point $X \in \mathbb{E}^3$ is therefore represented as
 - $X = (x_1, x_2, x_3)^T \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}$

- We add a 1-1 correspondance between $\mathbb{E}^3$ and $\mathbb{R}^3$
 - $X = (x_1 e_1 + x_2 e_2 + x_3 e_3) \in \mathbb{R}^3$



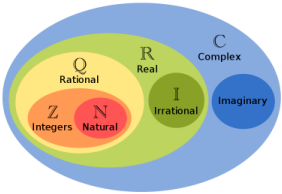
- Great ! We start to be able to measure some things between points of $\mathbb{E}^3$, by defining:
 - Similarity between two points (dot product)
 - New perpendicular points (cross product)
 - Angles between points (dot and cross products)
- But we still cannot measure lengths !

Summary

Add a law
 $+: G \times G \rightarrow G$

Add a law $\cdot: G \times G \rightarrow G$
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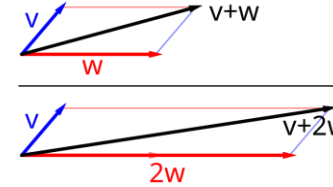


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Sets

Groups $(G, +)$

Field $(K, +, \cdot)$

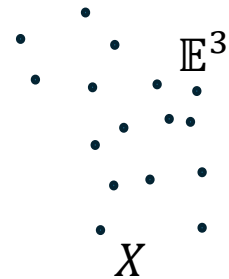
Vector space $(V, \oplus, \odot)$

Eléments = points

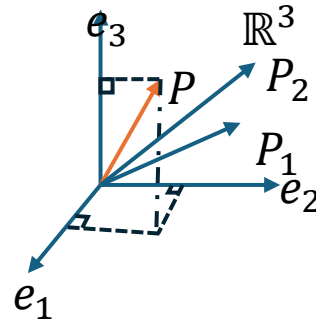
Add basis vectors
 Scalar product

Euclidean Space $\mathbb{E}^3$
 A set of points

3D Space $(\mathbb{R}^3, +, \cdot)$



Identify
 $\longleftrightarrow$



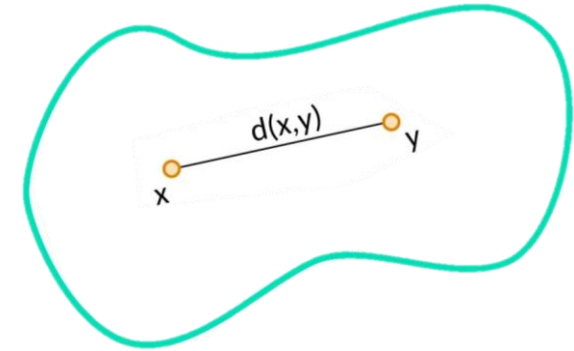
We have only topology

We cannot measure vector lengths, only projections

- We have now coordinates and projections of our points!
 And a vector space structure !

Euclidean Space $\mathbb{E}^3$ – Euclidean metric

- To measure distances we need a metric $d: V \times V \rightarrow K$
- The euclidean metric in $\mathbb{E}^3$ is **identified** to the inner product in $\mathbb{R}^3$:
 - $m(u, v) = u \cdot v = u^T v = u_1 v_1 + u_2 v_2 + u_3 v_3, \quad \forall u, v \in \mathbb{R}^3$
 - Using linear algebra it can be written without a dot as $u \cdot v = u^T v$
- With an euclidean metric, we can define vector lengths in $\mathbb{E}^3$
 - $\mu(v) = |v| = \sqrt{v \cdot v}, \quad \forall v \in \mathbb{R}^3$ **Why ?**
- *Definition*: the **Euclidean Space** $\mathbb{E}^3$



$\mathbb{E}^3$ is a space of points that can be identified with $\mathbb{R}^3$ and has a metric.

Three Dimensional Space $\mathbb{R}^3$ – Examples

- Let $\mathbf{v} = (3,0,0)^T$, $\mathbf{w} = (2,2,0)^T$
- $\mathbf{v} \cdot \mathbf{w} = 3 \cdot 2 + 0 \cdot 2 + 0 \cdot 0 = 6$
- What is the angle α between $\mathbf{v}$ and $\mathbf{w}$?
 - $|\mathbf{v}| = 3$, $|\mathbf{w}| = \sqrt{4 + 4 + 0} = 2\sqrt{2}$
 - So, $\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}||\mathbf{w}| \cos \alpha \Rightarrow \alpha = \arccos \frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}||\mathbf{w}|} = \arccos \frac{6}{3 \cdot 2\sqrt{2}} = \arccos \frac{\sqrt{2}}{2} = \frac{\pi}{4}$
- $\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}||\mathbf{w}| \cdot \cos \alpha = 3 \cdot 2\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 6$
- $\mathbf{v} \times \mathbf{w} = \begin{vmatrix} e_1 & e_2 & e_3 \\ 3 & 0 & 0 \\ 2 & 2 & 0 \end{vmatrix} = 0e_1 + 0e_2 + 6e_3$
- $|\mathbf{v} \times \mathbf{w}| = |\mathbf{v}||\mathbf{w}| \sin \alpha = \left| 3 \cdot 2\sqrt{2} \cdot \frac{\sqrt{2}}{2} \right| = 6$

Three Dimensional Space $\mathbb{R}^3$ – Examples

- Let $\mathbf{v} = (1,3,2)^T$, $\mathbf{w} = (2,4,1)^T$
- $\mathbf{v} \cdot \mathbf{w} = 1 \cdot 2 + 3 \cdot 4 + 2 \cdot 1 = 16$
- What is the angle α between $\mathbf{v}$ and $\mathbf{w}$?
 - Check it yourself !

- $\mathbf{v} \times \mathbf{w} = \begin{vmatrix} e_1 & e_2 & e_3 \\ 1 & 3 & 2 \\ 2 & 4 & 1 \end{vmatrix}$
 $= (3 \cdot 1 - 4 \cdot 2)e_1 + (2 \cdot 2 - 1 \cdot 1)e_2 + (1 \cdot 4 - 2 \cdot 3)e_3$
 $= -5e_1 + 3e_2 - 2e_3$

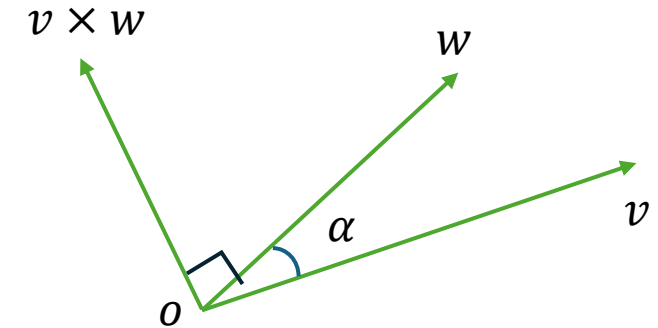
- $|\mathbf{v} \times \mathbf{w}| = \text{check it yourself !}$

Euclidean Space $\mathbb{E}^3$ – Cross-product

- We can also inherit the cross-product from $\mathbb{R}^3$

- $u \times v = \begin{vmatrix} e_1 & e_2 & e_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix} \in \mathbb{R}^3$

- The cross-product generates a vector normal to u and v in $\mathbb{E}^3$:
 - $(u \times v) \cdot u = (u \times v) \cdot v = 0$
 - $u \times v = -v \times u$



- Often in computer graphics, we want to use a matrix multiplication instead of a cross product

- $[u]_{\times} = \begin{pmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{pmatrix}$

- Now we can write $u \times v = u_{\times} v$
- $[u]_{\times}$ is a skew-symmetric matrix, ie: $[u]_{\times}^T = -[u]_{\times}$

Euclidean Space $\mathbb{E}^3$ – Cross-product

- Example:

- Let $\mathbf{v} = (1, 3, 2)^T$, $\mathbf{w} = (2, 4, 1)^T$

$$[\mathbf{u}]_{\times} = \begin{pmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{pmatrix}$$

- $\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ 1 & 3 & 2 \\ 2 & 4 & 1 \end{vmatrix}$
 $= (3 \cdot 1 - 4 \cdot 2)\mathbf{e}_1 + (2 \cdot 2 - 1 \cdot 1)\mathbf{e}_2 + (1 \cdot 4 - 2 \cdot 3)\mathbf{e}_3$
 $= -5\mathbf{e}_1 + 3\mathbf{e}_2 - 2\mathbf{e}_3$

- Now, $[\mathbf{v}]_{\times} = \begin{pmatrix} 0 & -2 & 3 \\ 2 & 0 & -1 \\ -3 & 1 & 0 \end{pmatrix}$

- So we can write:

- $\mathbf{v} \times \mathbf{w} = [\mathbf{v}]_{\times} \mathbf{w} = \begin{pmatrix} 0 & -2 & 3 \\ 2 & 0 & -1 \\ -3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \cdot 2 - 2 \cdot 4 + 3 \cdot 1 \\ 2 \cdot 2 + 0 \cdot 4 - 1 \cdot 1 \\ -3 \cdot 2 + 1 \cdot 4 + 0 \cdot 1 \end{pmatrix} = \begin{pmatrix} -5 \\ 3 \\ -2 \end{pmatrix}$

Euclidean Space $\mathbb{E}^3$ – Projection vector-vector

- As $\mathbf{v} \cdot \mathbf{u} = |\mathbf{v}| |\mathbf{u}| \cos \alpha$, the **length** of a vector $\mathbf{v}$ projected onto the vector $\mathbf{u}$ is

- $\mu(\mathbf{v}) = |\mathbf{v}| \cos \alpha = \frac{\mathbf{v} \cdot \mathbf{u}}{|\mathbf{u}|}$

- However, this length is a scalar (no particular orientation in our space).

- To report this to $\mathbf{u}$, we generate a unitary vector in the direction of $\mathbf{u}$: $\hat{\mathbf{u}} = \frac{\mathbf{u}}{|\mathbf{u}|}$

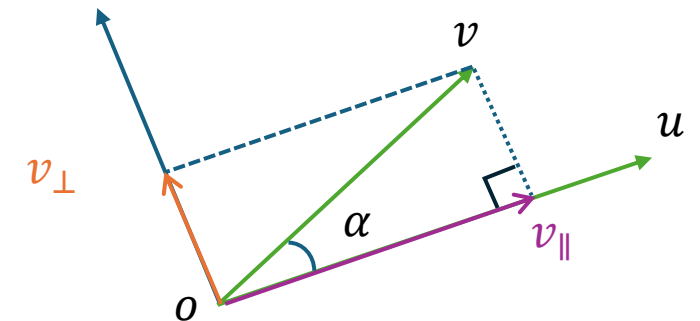
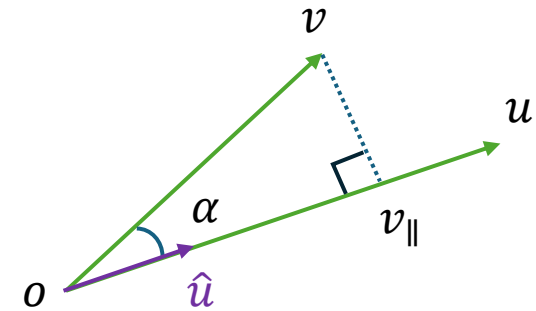
- The projection of $\mathbf{v}$ to $\mathbf{u}$ is

- $\text{proj}_{\mathbf{u}} \mathbf{v} = \mathbf{v}_{\parallel} = \frac{\mathbf{v} \cdot \mathbf{u}}{|\mathbf{u}|} \hat{\mathbf{u}} = \frac{\mathbf{v} \cdot \mathbf{u}}{|\mathbf{u}| |\mathbf{u}|} \mathbf{u} = \frac{\mathbf{v} \cdot \mathbf{u}}{|\mathbf{u}|^2} \mathbf{u}$

- In matrix version : $\text{proj}_{\mathbf{u}} \mathbf{v} = \frac{1}{|\mathbf{u}|^2} \begin{pmatrix} u_1^2 & u_1 u_2 & u_1 u_3 \\ u_1 u_2 & u_2^2 & u_2 u_3 \\ u_1 u_3 & u_2 u_3 & u_3^2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ (check it!)

- The perpendicular vector to $\mathbf{v}$ is

- $\text{perp}_{\mathbf{u}} \mathbf{v} = \mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel} = \mathbf{v} - \frac{\mathbf{v} \cdot \mathbf{u}}{|\mathbf{u}|^2} \mathbf{u}$



Normal vector – Example

- Example:

- Let $u = (2,1,2)^T$

- Then, $\hat{u} = \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)^T$

$$\hat{u} = \frac{u}{\|u\|}$$

- We want to reexpress u in the direction $\hat{u}$

- We have: $|u| = \sqrt{4 + 1 + 4} = 3$

- Then, $u = |u|\hat{u} = 3 \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix}$

Euclidean Space $\mathbb{E}^3$ – Projection vector-vector

- Example:

- Let $v = (1,4,2)^T$, $u = (2,1,2)^T$

- $v \cdot u = 1.2 + 4.1 + 2.2 = 10$

- $\hat{u} = \frac{u}{\|u\|} = \frac{(2,1,2)^T}{\sqrt{4+1+4}} = \frac{(2,1,2)^T}{3}$

- $|u| = \sqrt{2.2 + 1.1 + 2.2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$

- $proj_u v = v_{\parallel} = \frac{10}{3} \frac{(2,1,2)^T}{3}$

- $perp_u v = v_{\perp} = v - v_{\parallel} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} - \frac{10}{9} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$

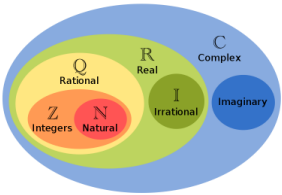
$$\hat{u} = \frac{u}{\|u\|}$$
$$proj_u v = v_{\parallel} = \frac{vu}{|u|} \hat{u} = \frac{vu}{|u|} \frac{u}{|u|} = \frac{vu}{\|u\|^2} u$$
$$perp_u v = v_{\perp} = v - v_{\parallel}$$

Summary

Add a law
 $+: G \times G \rightarrow G$

Add a law $\cdot: G \times G \rightarrow G$
 And $\cdot$ distribute $+$.

Objects = vectors
 Add 2 laws + inherit field ops
 $\oplus: V \times V \rightarrow V$
 $\odot: K \times V \rightarrow V$

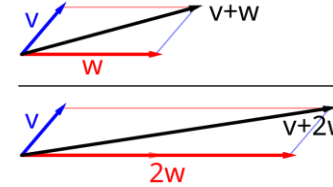


+	0	1	2	3
0	0	1	2	3
1	1	0	3	2
2	2	3	0	1
3	3	2	1	0

Addition Table

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	4	6
3	0	3	6	5

Multiplication Table



Sets

Groups $(G, .)$

Field $(K, +, .)$

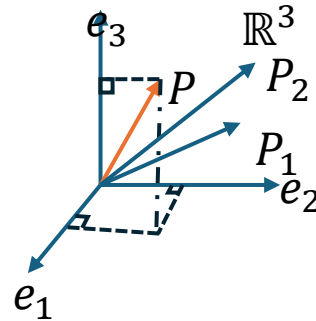
Vector space $(V, \oplus, \odot)$

Eléments = points

Add basis vectors
 Scalar product

Euclidean Space $\mathbb{E}^3$
 A set of points

3D Space $(\mathbb{R}^3, +, .)$



Identify
 $\longleftrightarrow$

Dot product
 cross-product

$d(v_1, v_2)$

$\mu(v)$

$proj_Q(v)$

We have only topology

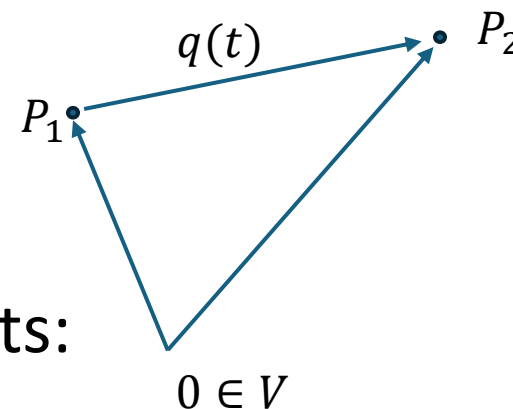
We cannot measure

We are stuck to origin

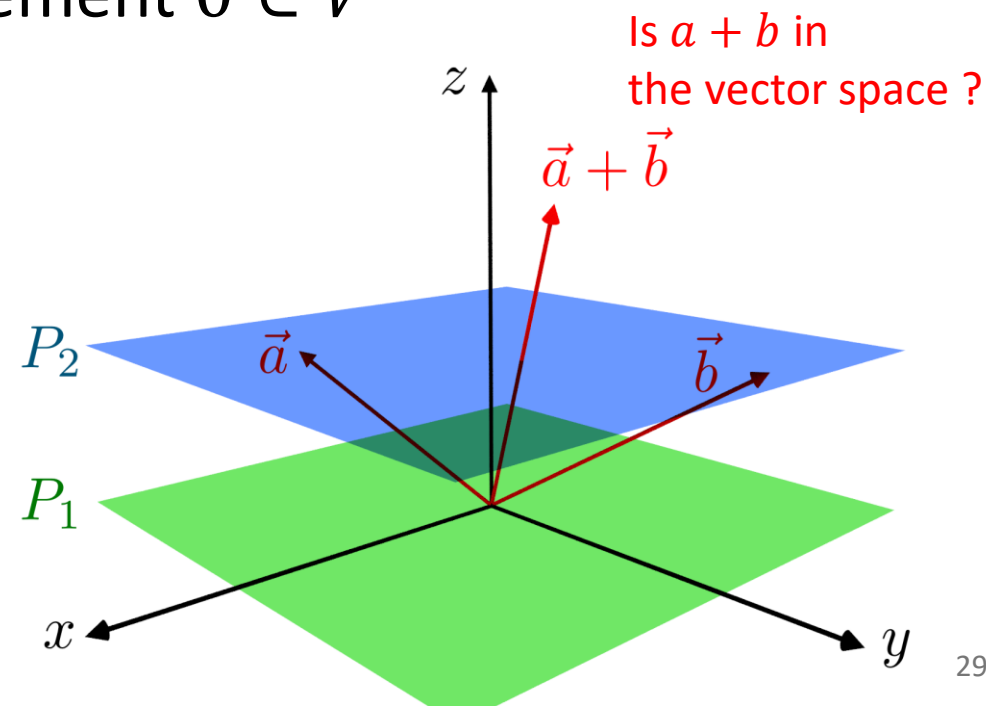
- We have defined distances, angles, etc. on a vector space ! Are all objects there?

Affine space

- Now we have: points, distances, projections, angles..
- We want to define lines (and other mathematical objects)
- Intuitively a line is a linear interpolation between two points:
 - $q(t) = (1 - t)P_1 + tP_2 \quad t \in [0, 1] \quad P_1, P_2 \in V$
- Technically, we have defined the maps $\cdot$ and $+$, so we can define q
- But.. A vector space is tied to an origin, the element $0 \in V$

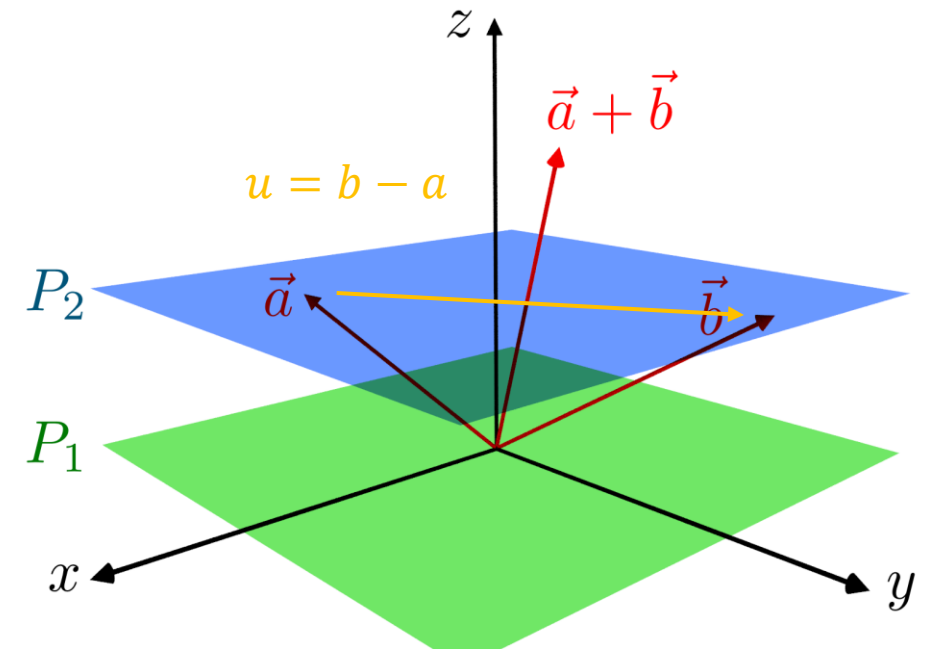


- Our line lives in a different space,
 - as it has not the same origin !
- *Informal*: An **affine space** is when we *forget* about the origin.



Affine space

- As another common example to define a vector which is in the affine space:
- Let $a, b \in V$, then $u = b - a$ is a vector but it does not necessarily start at the origin $0 \in V$!
- Vectors not starting at the origin are called *free-vectors*.
- Other example:
 - Solutions set of $Ax = b$ Why ?



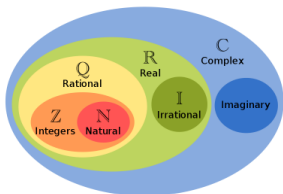
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Forget the origin $0 \in V$

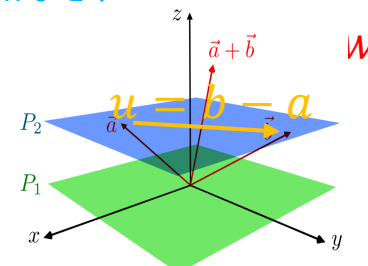
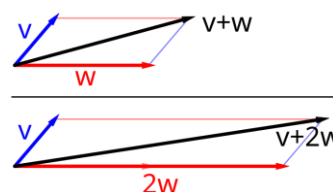


+	0	1	2	3
0	0	1	2	3
1	1	0	3	2
2	2	3	0	1
3	3	2	1	0

Addition Table

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	4	6
3	0	3	6	5

Multiplication Table



Sets

Groups $(G, \cdot)$

Field $(K, +, \cdot)$

Vector space $(V, \oplus, \odot)$

Affine spaces

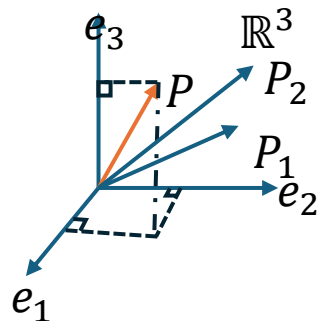
Eléments = points

Add basis vectors
 Scalar product

Remove origin

Euclidean Space $\mathbb{E}^3$
 A set of points

3D Space $(\mathbb{R}^3, +, \cdot)$



Dot product
 cross-product

$d(v_1, v_2)$

$\mu(v)$

$proj_Q(v)$

Identify

We have only topology

We cannot measure

We are stuck to origin

We have no objects in this space

- The affine space does not contain yet new objects and their operations

Euclidean Space $\mathbb{E}^3$ – Distance point-line

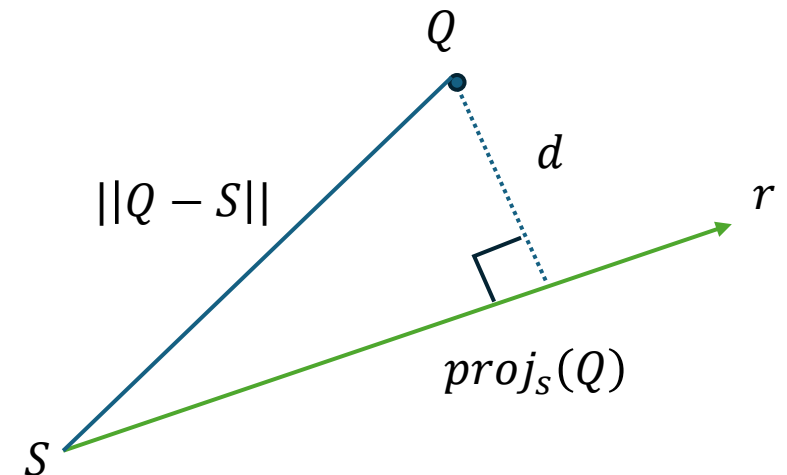
- Lines can have other representations:
 - $\forall U, V \in \mathbb{R}^3: q(t) = (1 - t)U + tV, \quad t \in [0, 1]$
 - A ray has a single starting point and expands from it $\forall S, K \in \mathbb{R}^3: r(t) = S + tK, \quad t \in \mathbb{R}$
 - Link between the two: $S = U, \quad K = V - U$
- Now that we have points, angles, projections, lines, etc.. We want to define some basic operations:

1. Distance point line

- Using Pythagorean theorem, the squared distance between a point Q and a line $r(t)$:

- $d^2 = (Q - S)^2 - \left(\text{proj}_Q(S) \right)^2 = (Q - S)^2 - \left(\frac{(Q-S)V}{V^2} V \right)^2$

- $d = \sqrt{(Q - S)^2 - \left(\frac{(Q-S)V}{V^2} V \right)^2}$



Euclidean Space $\mathbb{E}^3$ – Distance point-line

- Example:

- Let $r(t) = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ with $S = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, and a point $Q = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix}$

- We have:

- $Q - S = \begin{pmatrix} 5-3 \\ 2-1 \\ 3-2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

$$d = \sqrt{(Q - S)^2 - \left(\frac{(Q - S) \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \|\mathbf{v}\| \right)^2}$$

- $d^2 = (4 + 1 + 1) - \left(\frac{(2,1,1)^T (1,2,1)^T}{(1+4+1)} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right)^2 = 6 - \left(\frac{5}{6} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right)^2 = 6 - \left(\frac{25}{36} + \frac{100}{36} + \frac{25}{36} \right) = \frac{36}{6} - \frac{25}{6} = \frac{11}{6}$

- $d = \sqrt{\frac{11}{6}}$

Euclidean Space $\mathbb{E}^3$ – Distance line-line

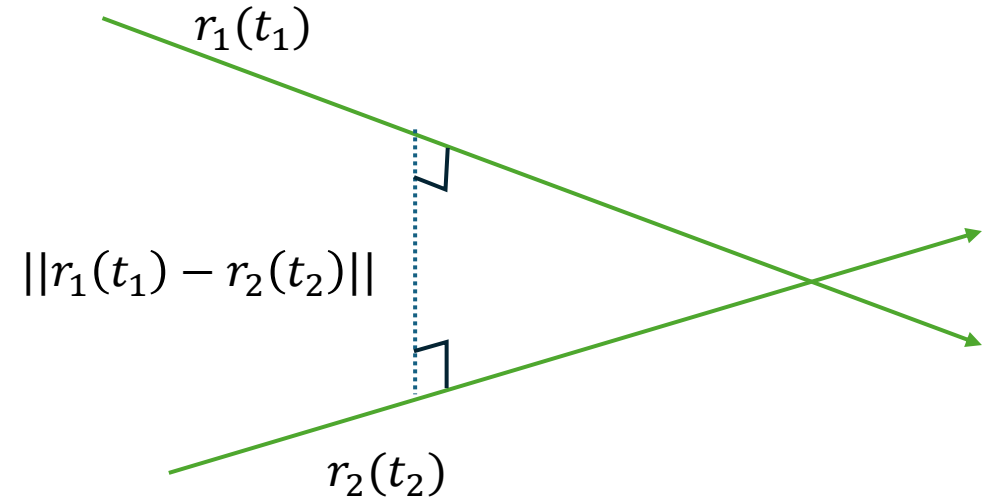
2. We can also find the distance between 2 lines

- Let $r_1(t_1) = S_1 + t_1V_1$, $r_2(t_2) = S_2 + t_2V_2$
be two lines.

- The distance between two points (t_1, t_2)
on the lines is:

- $$\begin{aligned} f(t_1, t_2) &= \|r_1(t_1) - r_2(t_2)\|^2 \\ &= r_1(t_1)^2 + r_2(t_2)^2 - 2r_1(t_1)r_2(t_2) \\ &= (S_1 + t_1V_1)^2 + (S_2 + t_2V_2)^2 - 2(S_1S_2 + t_1V_1S_2 + t_2V_2S_1 + t_1t_2V_1V_2) \\ &= |S_1|^2 + t_1^2|V_1|^2 + 2t_1S_1V_1 + |S_2|^2 + t_2^2|V_2|^2 + 2t_2S_2V_2 - 2(S_1S_2 + t_1V_1S_2 + t_2V_2S_1 + t_1t_2V_1V_2) \end{aligned}$$

Which product is this?



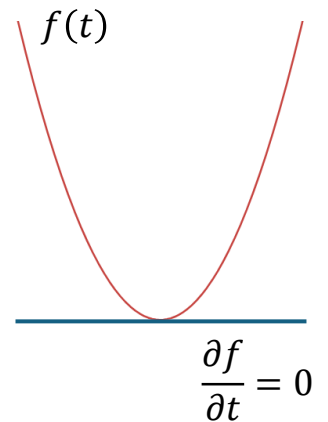
- We still need to find the minimal distance (so finding optimal t_1 and t_2)
- But 2 lines do not necessarily intersect in 3D, how to find this distance ?

Euclidean Space $\mathbb{E}^3$ – Distance line-line

- $f(t_1, t_2) = |S_1|^2 + t_1^2|V_1|^2 + 2t_1S_1V_1 + |S_2|^2 + t_2^2|V_2|^2 + 2t_2S_2V_2 - 2(S_1S_2 + t_1V_1S_2 + t_2V_2S_1 + t_1t_2V_1V_2)$

- We want to find the minimal value of f , so derivatives set to zero !

- $$\begin{cases} \frac{\partial f}{\partial t_1} = 2t_1V_1^2 + 2S_1V_1 - 2V_1S_2 - 2t_2V_1V_2 = 0 \\ \frac{\partial f}{\partial t_2} = 2t_2V_2^2 + 2S_2V_2 - 2V_2S_1 - 2t_1V_1V_2 = 0 \end{cases}$$



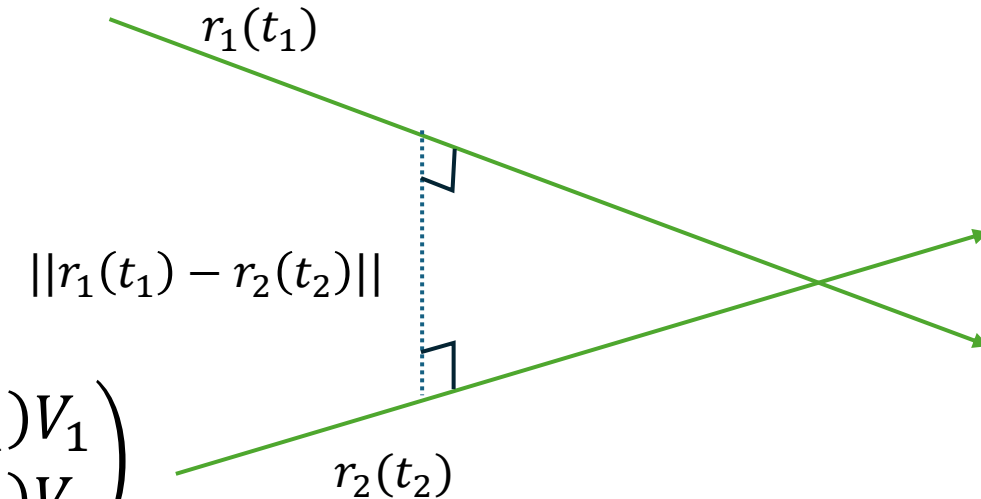
- This can be rewritten in matrix form
$$\begin{pmatrix} V_1^2 & -V_1V_2 \\ V_1V_2 & -V_2^2 \end{pmatrix} \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} = \begin{pmatrix} (S_2 - S_1)V_1 \\ (S_2 - S_1)V_2 \end{pmatrix}$$

Euclidean Space $\mathbb{E}^3$ – Distance line-line

- $f(t_1, t_2) = |S_1|^2 + t_1^2|V_1|^2 + 2t_1S_1V_1 + |S_2|^2 + t_2^2|V_2|^2 + 2t_2S_2V_2 - 2(S_1S_2 + t_1V_1S_2 + t_2V_2S_1 + t_1t_2V_1V_2)$
- Solving for t_1 and t_2 gives

$$\bullet \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} = \begin{pmatrix} |V_1|^2 & -V_1V_2 \\ V_1V_2 & -|V_2|^2 \end{pmatrix}^{-1} \begin{pmatrix} (S_2 - S_1)V_1 \\ (S_2 - S_1)V_2 \end{pmatrix}$$

$$\bullet \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} = \frac{1}{(V_1 \cdot V_2)^2 - |V_1|^2|V_2|^2} \begin{pmatrix} -|V_2|^2 & V_1V_2 \\ -V_1V_2 & |V_1|^2 \end{pmatrix} \begin{pmatrix} (S_2 - S_1)V_1 \\ (S_2 - S_1)V_2 \end{pmatrix}$$



- The distance is $d(P_1(t), P_2(t)) = \sqrt{f(t_1, t_2)}$
- If $(V_1V_2)^2 - |V_1|^2|V_2|^2 = 0$, then the lines are parallel.

Euclidean Space $\mathbb{E}^3$ – Distance line-line

- Example:

- Let $S_1 = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}, V_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix},$

- then $r_1(t_1) = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} + t_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad |S_1|^2 = 9 + 1 + 4 = 14, \quad |V_1|^2 = 1 + 4 + 1 = 6$

- Let $S_2 = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}, V_2 = \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix},$

- then $r_2(t_2) = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} + t_2 \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}, \quad |S_2|^2 = 4 + 4 + 1 = 9, \quad |V_2|^2 = 0 + 1 + 25 = 26$

- $$\begin{aligned} f(t_1, t_2) &= 14 + 6 t_1^2 + 2 t_1 (3 \cdot 1 + 1 \cdot 2 + 2 \cdot 1) + 9 + 26 t_2^2 + 2 t_2 (-2 \cdot 0 + 2 \cdot 1 + 1 \cdot 5) \\ &\quad - 2[(3 \cdot (-2) + 1 \cdot 2 + 2 \cdot 1) + t_1 (1 \cdot (-2) + 2 \cdot 2 + 1 \cdot 1) + t_2 (0 \cdot 3 + 1 \cdot 1 + 5 \cdot 2) + t_1 t_2 (1 \cdot 0 + 2 \cdot 1 + 1 \cdot 5)] \\ &= 14 + 6 t_1^2 + 14 t_1 + 9 + 26 t_2^2 + 14 t_2 - 2[-2 + 3 t_1 + 11 t_2 + 7 t_1 t_2] \\ &= 27 + 6 t_1^2 + 8 t_1 + 26 t_2^2 - 8 t_2 - 14 t_1 t_2 \end{aligned}$$

- $d(P_1(t_1), P_2(t_2)) = \sqrt{27 + 6 t_1^2 + 8 t_1 + 26 t_2^2 - 8 t_2 - 14 t_1 t_2}$

$$\begin{aligned} f(t_1, t_2) &= |S_1|^2 + t_1^2 |V_1|^2 + 2 t_1 S_1 V_1 + |S_2|^2 + t_2^2 |V_2|^2 + 2 t_2 S_2 V_2 \\ &\quad - 2(S_1 S_2 + t_1 V_1 S_2 + t_2 V_2 S_1 + t_1 t_2 V_1 V_2) \end{aligned}$$

Euclidean Space $\mathbb{E}^3$ – Distance line-line

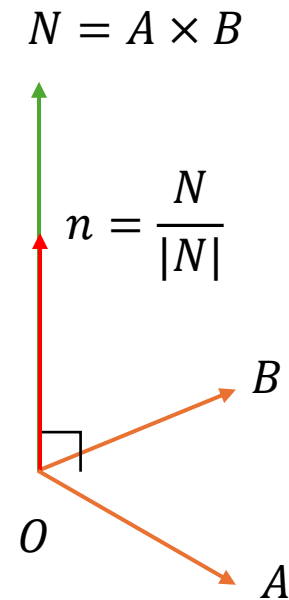
- Let $S_1 = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$, $V_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$,
 - then $r_1(t_1) = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} + t_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $|S_1|^2 = 9 + 1 + 4 = 14$, $|V_1|^2 = 1 + 4 + 1 = 6$
- Let $S_2 = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$, $V_2 = \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}$,
 - then $r_2(t_2) = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} + t_2 \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}$, $|S_2|^2 = 4 + 4 + 1 = 9$, $|V_2|^2 = 0 + 1 + 25 = 26$
- We still need to find the values of t_1 and t_2 in $d(P_1(t_1), P_2(t_2)) = \sqrt{27 + 6 t_2^2 + 8 t_1 + 26 t_2^2 - 8 t_2 - 14 t_1 t_2}$
- $$\begin{pmatrix} t_1 \\ t_2 \end{pmatrix} = \frac{1}{(V_1 \cdot V_2)^2 - |V_1|^2 |V_2|^2} \begin{pmatrix} -|V_2|^2 & V_1 V_2 \\ -V_1 V_2 & |V_1|^2 \end{pmatrix} \begin{pmatrix} (S_2 - S_1) V_1 \\ (S_2 - S_1) V_2 \end{pmatrix} =$$

$$\frac{1}{((1,2,1)^T \cdot (0,1,5)^T)^2 - 6 \cdot 26} \begin{pmatrix} -26 & (1,2,1)^T (0,1,5)^T \\ -(1,2,1)^T (0,1,5)^T & 6 \end{pmatrix} \begin{pmatrix} ((-2,2,1)^T - (3,1,2)^T) (1,2,1)^T \\ ((-2,2,1)^T - (3,1,2)^T) (0,1,5)^T \end{pmatrix} =$$

$$\frac{1}{(7)^2 - 6 \cdot 26} \begin{pmatrix} -26 & 7 \\ -7 & 6 \end{pmatrix} \begin{pmatrix} (-5,1,-1)^T (1,2,1)^T \\ (-5,1,-1)^T (0,1,5)^T \end{pmatrix} = \frac{1}{49 - 156} \begin{pmatrix} -26 & 7 \\ -7 & 6 \end{pmatrix} \begin{pmatrix} -4 \\ -4 \end{pmatrix} = \frac{1}{-107} \begin{pmatrix} 76 \\ 4 \end{pmatrix}$$
- By plugging those values back in d, we find the final distance !

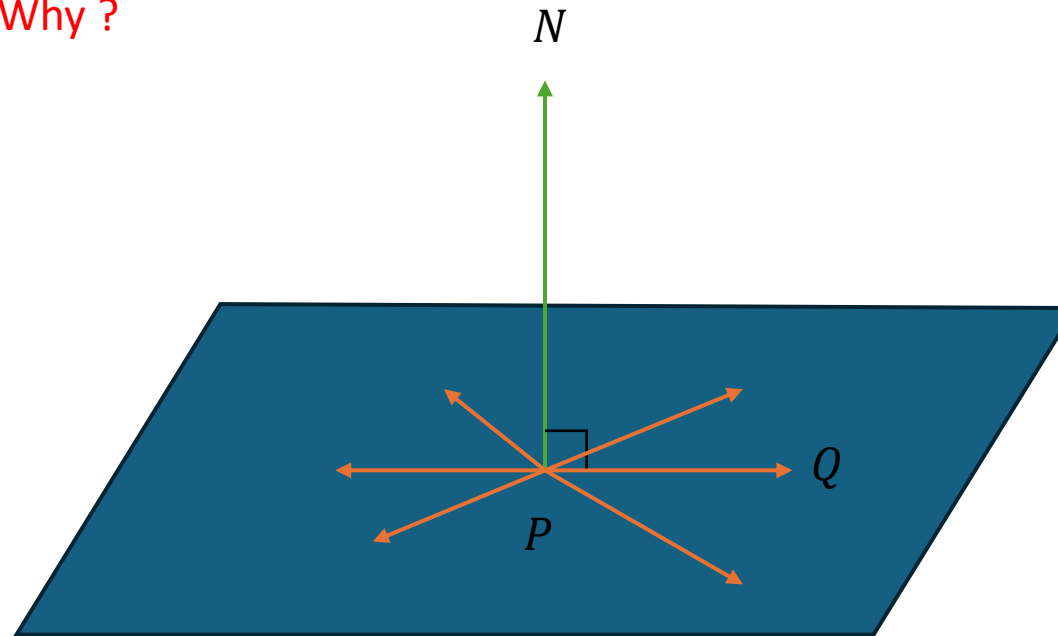
Euclidean Space $\mathbb{E}^3$

- We have lines, what about planes ?
- To make planes, we need the notion of a vector normal to a surface:
 - Suppose we have two vectors A and B with a shared origin.
 - They implicitly define a surface
 - Their normal is defined as $N = A \times B$
 - The unit normal is given by $n = \frac{N}{|N|}$



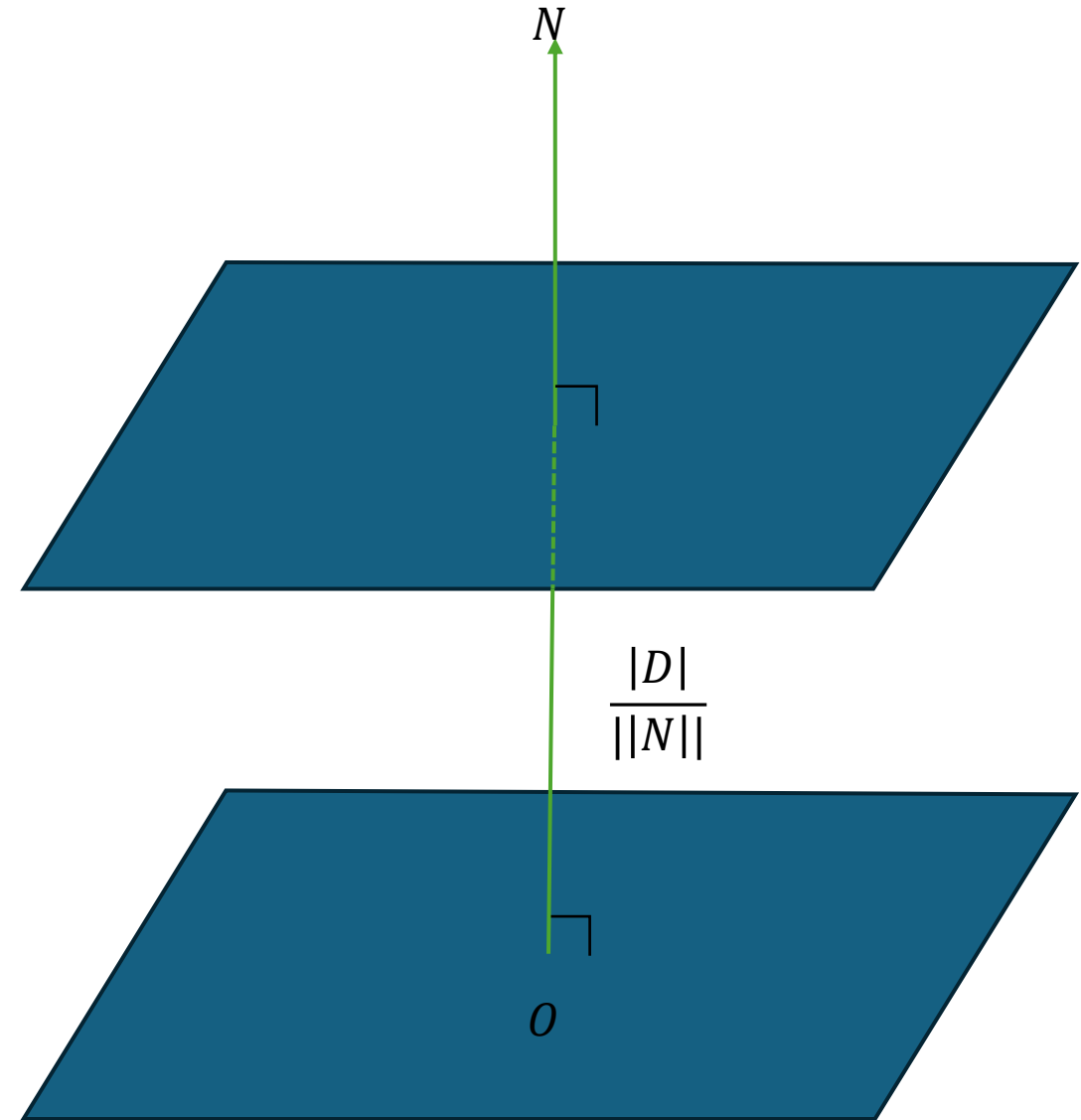
Euclidean Space $\mathbb{E}^3$ – Planes

- It's time to define a **plane**:
- Given a 3D point P and a normal vector $N = (A, B, C)^T$, the plane π passing through the point P and perpendicular to the direction N is defined as the set of points Q such that $\pi \equiv N \cdot (Q - P) = 0$.
- This is commonly written as $\pi \equiv Ax + By + Cz + D = 0$ Why ?
- With $D = -N \cdot P$ Why ?



Euclidean Space $\mathbb{E}^3$ – Planes

- Summary:
- $\pi \equiv Ax + By + Cz + D = 0$
- With $D = -N \cdot P$
- $\frac{|D|}{||N||}$ is the distance by which the plane is offset from a parallel plane that passes through the origin.



Is this plane in the vector space ?

Euclidean Space $\mathbb{E}^3$ – Planes

- Example:
- Suppose we have a point $P = (-4, 2, 2)^T$ and a normal vector $n = (3, 9, 1)^T$
- We have:

- $\pi \equiv n \cdot (Q - P) = 0$

- $\pi \equiv \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} -4 \\ 2 \\ 2 \end{pmatrix} \right) = 0$

- $\Leftrightarrow \pi \equiv \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x+4 \\ y-2 \\ z-2 \end{pmatrix} = 3(x+4) + 9(y-2) + (z-2) = 0$

- $\pi \equiv 3x + 12 + 9y - 18 + z - 2 = 0$

- $\pi \equiv 3x + 9y + z - 8 = 0$

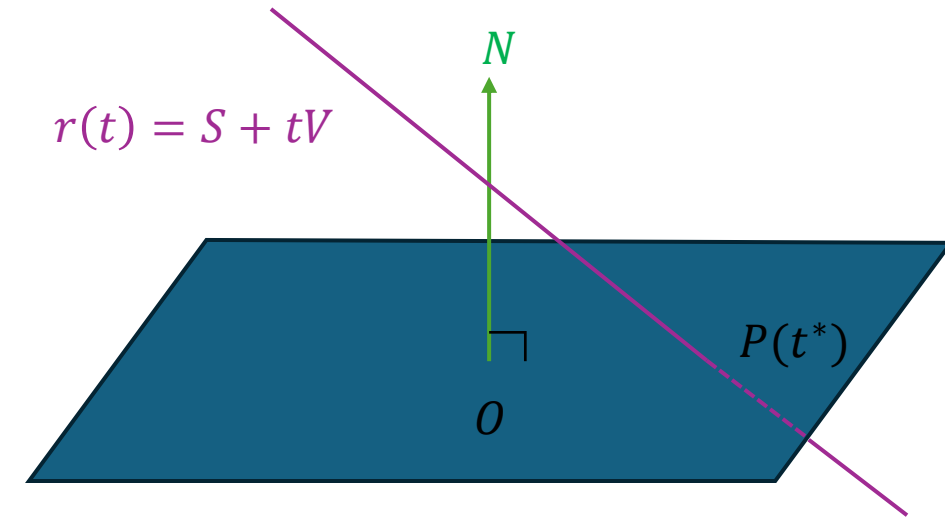
- Offset distance: $\frac{|D|}{||n||} = \frac{|-8|}{\sqrt{10}} = \frac{8}{\sqrt{10}}$

Euclidean Space $\mathbb{E}^3$ – Intersection Plane-Line

- Let's continue to define operations:

3. Intersection of a line and a plane in 3D

- Plane: $\pi \equiv N \cdot (Q - P) = N \cdot Q + D = 0$.
- The line: $r(t) = S + tV$



- We want the t^* both on the plane and the line.

- $\pi \equiv N \cdot r(t^*) + D = 0 \Rightarrow N \cdot (S + t^*V) + D = 0 \Leftrightarrow t^* = \frac{-(N \cdot S + D)}{N \cdot V}$
Why?

- Plugging back the t^* in the line equation $r(t^*)$, we obtain the point of intersection

- $r(t^*) = S + \frac{-(N \cdot S + D)}{N \cdot V} V$

- If $N \cdot V = 0$, then the line and the plane are parallel.
 - If further $N \cdot V + D = 0$, then the line is inside the plane.

Euclidean Space $\mathbb{E}^3$ – Intersection Plane-Line

- Line: $l \equiv \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 - t \\ 1 + t \\ 3t \end{pmatrix}$
- Plane: $\pi \equiv 3x - 2y + z - 10 = 0$
- Intersection:
 - $\pi \equiv 3(2 - t) - 2(1 + t) + 3t - 10 = 0$
 - $\Leftrightarrow 6 - 3t - 2 - 2t + 3t - 10 = -6 - 2t = 0$
 - $\Leftrightarrow t = -\frac{6}{2} = -3$
- Plugging it back to the line equation to find the point:
- $p = \begin{pmatrix} 2 - t \\ 1 + t \\ 3t \end{pmatrix} = \begin{pmatrix} 2 + 3 \\ 1 - 3 \\ -9 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ -9 \end{pmatrix}$
- How could you use the formulas instead ?

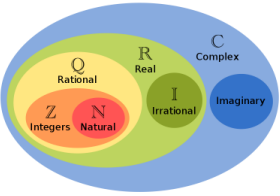
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Add a law $\cdot: G \times G \rightarrow G$
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 $\oplus: V \times V \rightarrow V$
 $\odot: K \times V \rightarrow V$

Forget the origin $0 \in V$

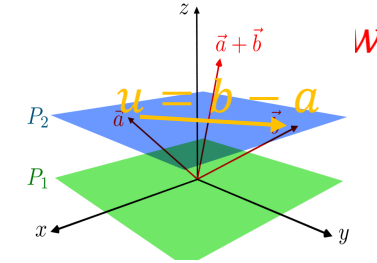
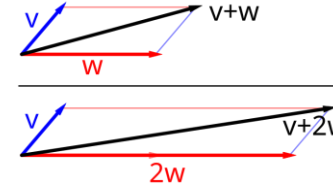


+	0	1	2	3
0	0	1	2	3
1	1	0	3	2
2	2	3	0	1
3	3	2	1	0

Addition Table

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	4	6
3	0	3	6	5

Multiplication Table



Sets

Groups $(G, \cdot)$

Field $(K, +, \cdot)$

Vector space $(V, \oplus, \odot)$

Affine spaces

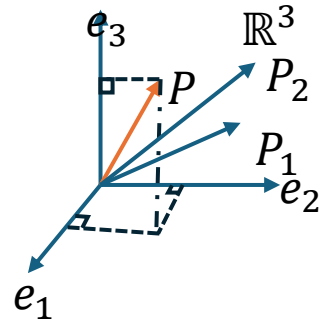
Eléments = points

Add basis vectors
 Scalar product

Remove origin

Euclidean Space $\mathbb{E}^3$
 A set of points

3D Space $(\mathbb{R}^3, +, \cdot)$



Dot product
 cross-product

$d(v_1, v_2)$

$\mu(v)$

$proj_Q(v)$

lines

Ops: point-line, line-line

Planes

Ops: line-plane, normal

Identify

We have only topology

We cannot measure

We are stuck to origin

We have no movements

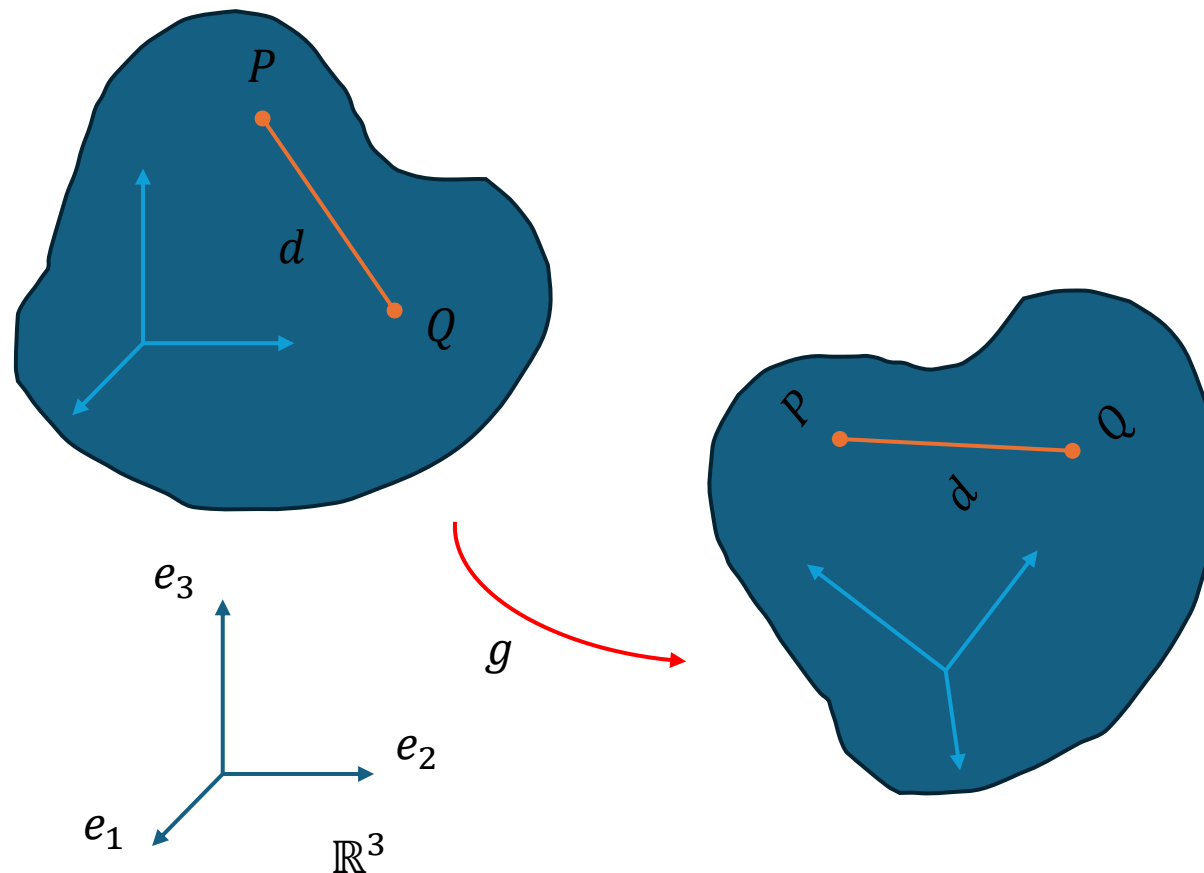
- We cannot move easily a set of points in space.

From points to movements

- We have defined:
 - Points
 - Operations
 - Distances
 - Lengths
 - Angles
 - Lines
 - Planes
 - Operations on lines and planes
- Now, we want to be able to move a set of points in space.
- The movement should have translations, but also rotations !

Rigid-body motion $SE(3)$

- When a body moves, we have to specify the movement for each material point inside
- In a rigid-body, the distance between any two points does not change with time!
- For rigid-bodies, only one point movement and the motion of the three coordinates axes attached to that point suffices.



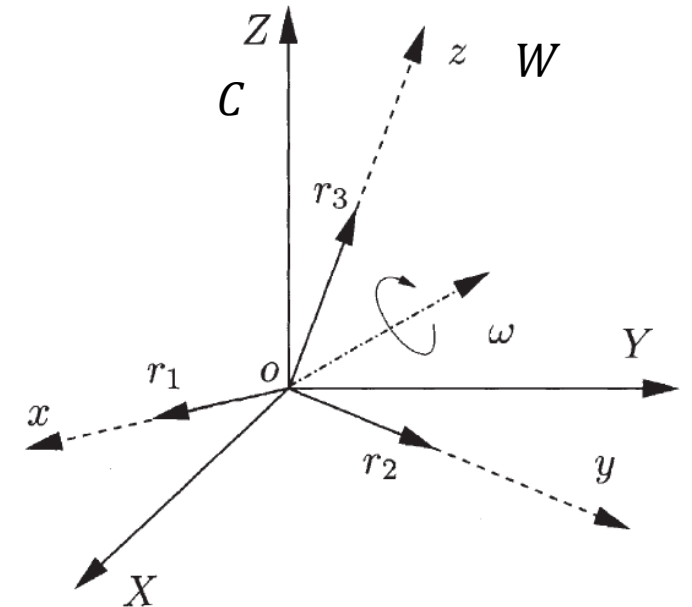
Rigid-body motion $SE(3)$

- Is any map, preserving the distance in $\mathbb{E}^3$, a rigid body motion ?
 - No, look at $f: (x_1, x_2, x_3)^T \mapsto (x_1, x_2, -x_3)^T$ (distance preserved, not orientation)
- Need a map $g(t): \mathbb{R}^3 \rightarrow \mathbb{R}^3: X \mapsto g(t)(X)$ preserving point's distances and orientation.
 - g must preserve vectors, if $v = Y - X$, then $g_*(v) = g(Y) - g(X)$
 - Independently of time, we can define it between starting and ending points:
 - $g: \mathbb{R}^3 \rightarrow \mathbb{R}^3: X \mapsto g(X)$
- **Definition:** Rigid Body Motion or **Special Euclidean Transformation $SE(3)$:**
 - A map $g: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that preserves the norm and the cross-product of any two vectors:
 - $\|g_*(v)\| = \|v\|$
 - $g_*(u) \times g_*(v) = g_*(u \times v), \forall u, v \in \mathbb{R}^3$
- Remember: g represents some rotation and translation that we can apply to vectors.

Why both conditions?

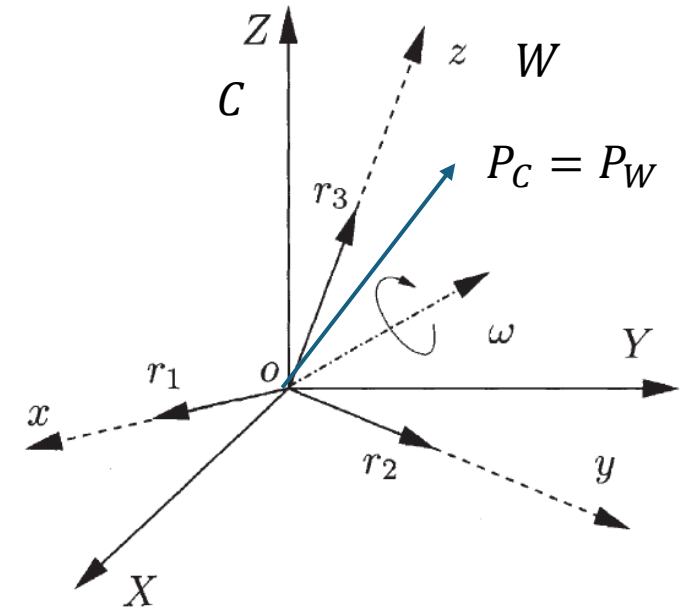
Rigid-body motion $SE(3)$ – Rotation

- Suppose a rigid object rotating about a fixed point $o \in \mathbb{E}^3$ and around an arbitrary axis ω in an frame of reference C on o .
- Suppose we have our mapping g as defined before.
- Let P_C be a point inside the rigid body in the same frame of reference.
- Instead of moving the point P_C , we can study how the reference frame C rotates with respect to the axis ω !
- Let's call this new frame W .
- The point P_C has coordinates (P_{1C}, P_{2C}, P_{3C})



Rigid-body motion SE(3) – Rotation

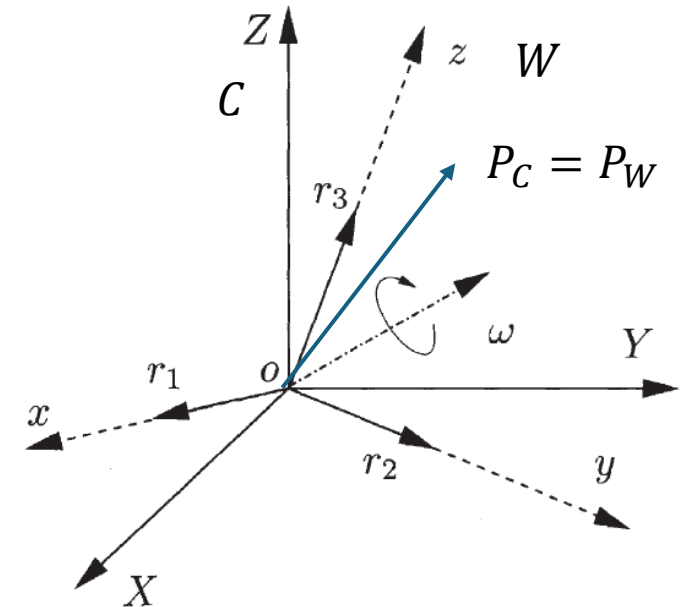
- Suppose a rigid object rotating about a fixed point $o \in \mathbb{E}^3$ and around an arbitrary axis ω in an frame of reference C on o .
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- Instead of moving the point P_C , we can study how the reference frame C rotates with respect to the axis ω !
- Let's call this new frame W .
- The point P_C has coordinates (P_{1C}, P_{2C}, P_{3C})
- In the frame W , it will be $P_W = (P_{1W}, P_{2W}, P_{3W})$
 - How to link the two representations ?



Rigid-body motion SE(3) – Rotation

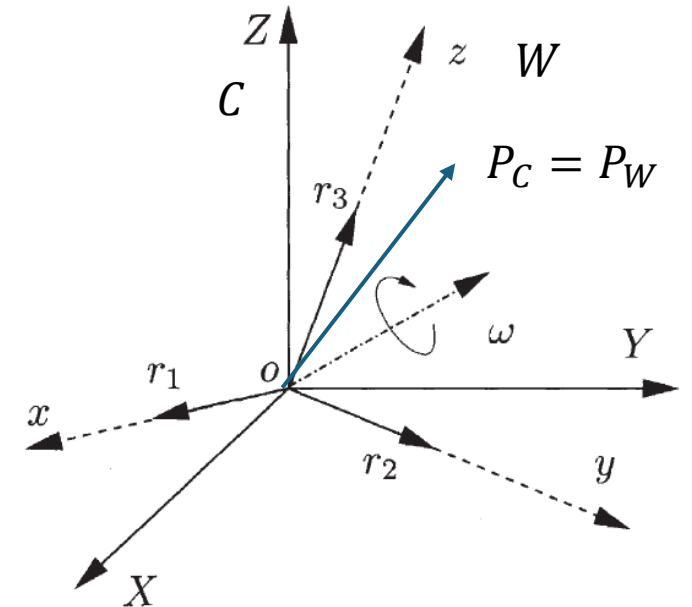
- The point P_C has coordinates (P_{1C}, P_{2C}, P_{3C})
- In the frame W , it will be $P_W = (P_{1W}, P_{2W}, P_{3W})$
 - How to link the two representations ?
- As points are determined by the basis vectors, we can study the effect of $g: C \rightarrow W$ on the basis of C .
 - $r_1 = g_*(e_1)$, $r_2 = g_*(e_2)$, $r_3 = g_*(e_3)$, $r_1, r_2, r_3 \in W$
- As $r_1, r_2, r_3 \in \mathbb{R}^3$, also form a basis in C , we can re-express P_C in that basis:

$$\begin{aligned} P_C &= P_{1W}r_1 + P_{2W}r_2 + P_{3W}r_3 = \begin{pmatrix} r_{1x} & r_{1y} & r_{1z} \\ r_{2x} & r_{2y} & r_{2z} \\ r_{3x} & r_{3y} & r_{3z} \end{pmatrix} \begin{pmatrix} P_{1W} \\ P_{2W} \\ P_{3W} \end{pmatrix} \\ &= R_{WC}P_W \end{aligned}$$



Rigid-body motion $SE(3)$ – Rotation

- So, we have basis elements for W
 - $r_1 = g_*(e_1)$, $r_2 = g_*(e_2)$, $r_3 = g_*(e_3)$, $r_1, r_2, r_3 \in W$
- And a conversion matrix from W to C using g :
 - $P_C = R_{WC}P_W$
- As g was a rotation, R_{WC} is a rotation matrix !
- Note 2 properties of R :
 - Since $r_i^T r_j = \delta_{ij}$, we have $R^T R = R R^T = I$
 - $\det(R) = r_1^T (r_2 \times r_3) = +1$ Check it!



Rotational Motion $SO(3)$

- For free, we have a way to get the inverse operation:
 - $P_W = R_{WC}^{-1}P_C = R_{WC}^T P_C = R_{CW}X_C$
- Composition law:
 - Two rotation matrices can be composed to obtain the final rotation, suppose we have a third coordinate frame K, then
 - $R_{KC} = R_{KW}R_{WC}$ Check it!
- *Definition*: Rotation matrices, or **Special Orthogonal group $SO(3)$** :
 - Special due to the fact that the $\det(R) = +1$ (orientation preserving)

$$SO(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \quad \det(R) = +1\}$$

Rotational Motion $SO(3)$ – Euler angles

- You have already seen several rotations in the past years
- They were **axis aligned** and called **Euler angles**
- Suppose a point is moved with angles θ, φ, γ around the x –, y –, z –axes
- $R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, R_y(\varphi) = \begin{pmatrix} \cos \varphi & 0 & \sin \varphi \\ 0 & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{pmatrix}, R_z(\gamma) = \begin{pmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- Those matrices allow to rotate the 3 Euler angles around the system axis.
- We can now check that rotation matrices using euler angles are in fact rotations !
- They all belongs to $SO(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \det(R) = +1\}$ (**check it**)
- An object rotating around the 3 axis uses the composition $R = R_z R_y R_x$
 - Is this composition commutative ? (**check it**)

Rotational Motion SO(3) – Euler angles

- $R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$, $R_y(\varphi) = \begin{pmatrix} \cos \varphi & 0 & \sin \varphi \\ 0 & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{pmatrix}$, $R_z(\gamma) = \begin{pmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- An object rotating around the 3 axis uses the composition $R = R_z R_y R_x$
- As an example, let's take $p = (1,2,3)^T$ and $\theta = \frac{\pi}{2}$ around x .
- $p' = R_x\left(\frac{\pi}{2}\right)p = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} \\ 0 & \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \cos \frac{\pi}{2} - 3 \sin \frac{\pi}{2} \\ 2 \sin \frac{\pi}{2} + 3 \cos \frac{\pi}{2} \end{pmatrix} = \begin{pmatrix} 1 \\ 2.0 - 3.1 \\ 2.1 + 3.0 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$
- We see that the x axis didn't change
- As the rotation was $\theta = \frac{\pi}{2}$, there is an inversion of the roles of 2 and 3
- Note the sign, **why it has changed ?**

Rotational Motion SO(3) – Euler angles

- Composing rotations can become pretty heavy very fast

- $R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, R_y(\varphi) = \begin{pmatrix} \cos \varphi & 0 & \sin \varphi \\ 0 & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{pmatrix}, R_z(\gamma) = \begin{pmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix}$

- E.g. : $R = R_z(\gamma)R_y(\varphi) = \begin{pmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \varphi & 0 & \sin \varphi \\ 0 & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{pmatrix} = \begin{pmatrix} \cos \gamma \cos \varphi & -\sin \gamma \cos \varphi & \sin \varphi \\ \sin \gamma \cos \varphi & \cos \gamma \cos \varphi & \sin \gamma \sin \varphi \\ -\sin \gamma \sin \varphi & \cos \gamma \sin \varphi & \cos \varphi \end{pmatrix}$

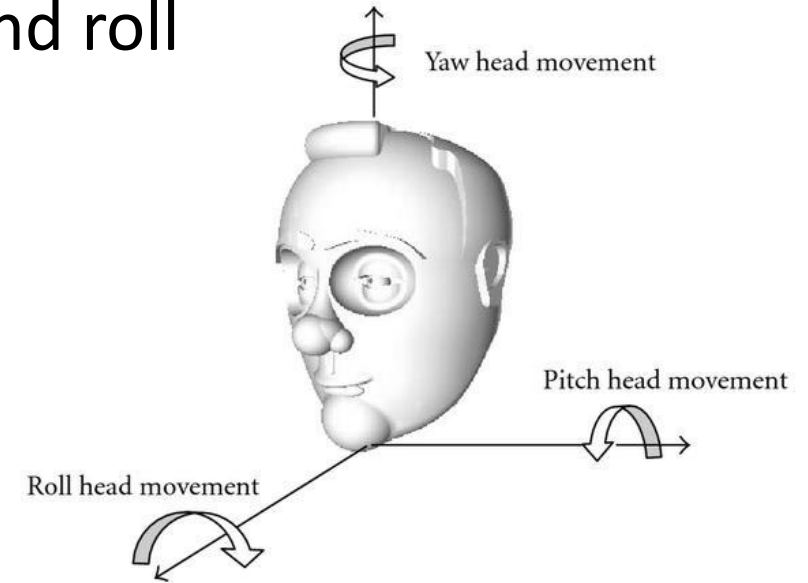
- There are many possible combinations:

- $R_z R_x, R_y R_x, R_x R_y R_z, \dots$

- Can be pretty heavy to compute compositions of rotations.

Rotational Motion $SO(3)$ – Euler angles

- Euler angles $(\theta, \varphi, \gamma)$ are often named yaw, pitch and roll
- They have two big limitations:
- **Gimbal lock:**
 - Occurs when two rotation axes align, reducing degrees of freedom from 3 to 2.
 - Example: Aircraft can no longer differentiate pitch from yaw at a specific orientation.
- Operation order dependency:
 - The order in which we multiply the rotation matrices changes the final rotation !
- We will see in the next lesson how to avoid those issues.



Rotational Motion $SO(3)$

- Do all rotation matrices look like the previous ones ?

$$SO(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \quad \det(R) = +1\}$$

- No, actually, **any** 3×3 matrix with $\det(R) = +1$ and $R^T R = I$ is a rotation !
 - But... Around an arbitrary axis.
- The previous ones ($R_x(\theta)$, $R_y(\varphi)$, $R_z(\gamma)$) are called **Euler rotations** and they represent rotations around the principal axis.
- Elements in $SO(3)$ are simply called **Rotation Matrices**.
- $\det(R) = +1$ and $R^T R = I$ are very useful. Sometimes you do not know what a matrix does in a software and you need to figure it out.
 - happens often in debugging robotics and computer vision.
- If you see a 3×3 matrix containing values, you only need to check $R^T R = I$ and its determinant! If its $+1$ then you automatically know that this matrix represent a rotation around some axis !

Rotational Motion $SO(3)$

- Two main limitations of rotation matrices:
- Over-parametrization:
 - 9 elements encode only 3 degrees of freedom. This redundancy complicates optimization problems.
- Numerical drift:
 - Repeated operations can lead to loss of orthogonality due to floating-point errors.
 - Regular re-orthogonalization is required.
- We will see in the next lesson how to do better.

Rigid-body motion $SE(3)$ – $R + T$

- Now that we have rotations
- We can add translations, and this defines the full map of rigid-body transformations
- A translation $T_{WC} \in \mathbb{R}^3$ simply translated a vector X_C from a frame C to a frame W , i.e., $X_W = X_C + T_{WC}$
- A rigid-body motion is then fully described as
 - $X_W = R_{WC}X_C + T_{WC}$
- *Definition (refined):* Rigid-body motion, or **Special Euclidean group $SE(3)$** :

$$SE(3) = \{ g = (R, T) \mid R \in SO(3), T \in \mathbb{R}^3 \}$$

Rigid-body motion $SE(3)$ – $R + T$ – Example

- *Example:*
- $SE(3) = \{ g = (R, T) \mid R \in SO(3), T \in \mathbb{R}^3 \}$
- Let $R = R_x\left(\frac{\pi}{2}\right)$, $T = (4, 5, 6)^T$, then $g = \left(R_x\left(\frac{\pi}{2}\right), \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \right)$
- Let $p = (1, 2, 3)^T$
- Applying g to p gives:
- $X = R_x\left(\frac{\pi}{2}\right) p + \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 8 \end{pmatrix}$
- The others points of a body containing p move the same way ! Per construction.

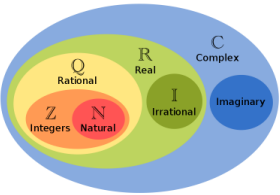
Summary

Add a law
 $+: G \times G \rightarrow G$

Add a law $\cdot: G \times G \rightarrow G$
 And $\cdot$ distribute $+$.

Objects = vectors
 Add 2 laws + inherit field ops
 $\oplus: V \times V \rightarrow V$
 $\odot: K \times V \rightarrow V$

Forget the origin $0 \in V$

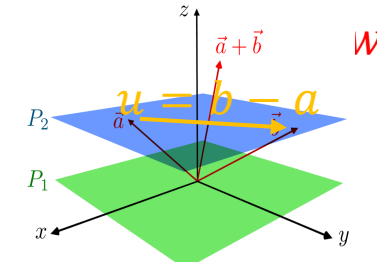
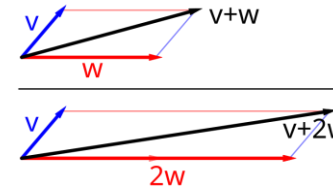


+	0	1	2	3
0	0	1	2	3
1	1	0	3	2
2	2	3	0	1
3	3	2	1	0

Addition Table

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	4	6
3	0	3	6	5

Multiplication Table



Sets

Groups $(G, \cdot)$

Field $(K, +, \cdot)$

Vector space $(V, \oplus, \odot)$

Affine spaces

Eléments = points

Add basis vectors
 Scalar product

Remove origin

Euclidean Space $\mathbb{E}^3$
 A set of points

3D Space $(\mathbb{R}^3, +, \cdot)$

lines

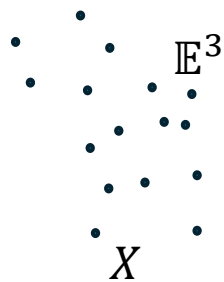
Ops: point-line, line-line

Planes

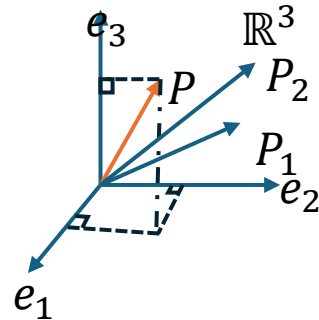
Ops: line-plane, normal

Rigid bodies $SE(3)$

Ops: (R, t)



Identify
 $\longleftrightarrow$



Dot product
 cross-product

$d(v_1, v_2)$

$\mu(v)$

$proj_Q(v)$

We have only topology

We cannot measure

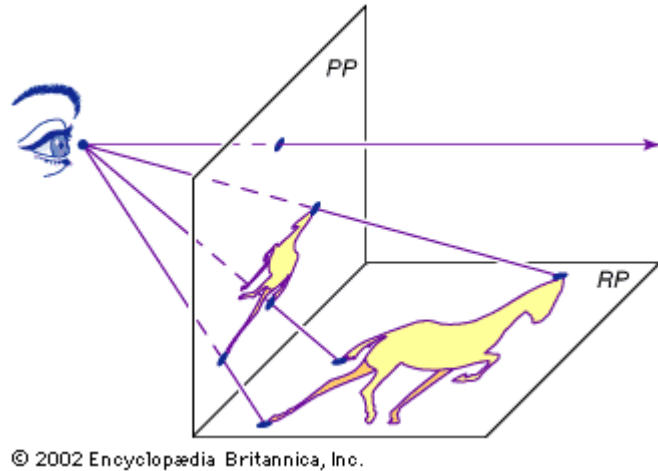
We are stuck to origin

We have no matrices

- Our rigid transform is not in a matrix form $g = (R, T)$

Next time

- We continue to build the needed structures!
- In particular: Projective Geometry



- And we revisit $SO(3)$ and other kinds of rotations !

Appendix: Skew symmetric matrices

- *Definition:* Skew-symmetric matrix

A matrix $M \in \mathbb{R}^{3 \times 3}$ is skew-symmetric iff $M = [u]_{\times}$ for some $u \in \mathbb{R}^3$

- Therefore $\mathbb{R}^3$ and the space of all skew-symmetric matrices are isomorphic.
- the $\times$ operator
 - $[]_{\times}: \mathbb{R}^3 \rightarrow so(3)$
- Its inverse is the v operator
 - $[]_v: so(3) \rightarrow \mathbb{R}^3$

Appendix: SE(3) and preservation of angles

- Why the rigid body motion preserves the angle between vectors ?
- Rigid body motion definition:
 - A map $g(t): \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that preserves the norm and the cross-product of any two vectors:
 - $||g_*(v)|| = ||v||$
 - $g_*(u) \times g_*(v) = g_*(u \times v), \forall u, v \in \mathbb{R}^3$
- The polarizing identity expresses the dot product in terms of the norm:
 - $u \cdot v = \frac{1}{4} (||u + v||^2 - ||u - v||^2)$
- And since $||u + v|| = ||g_*(u) + g_*(v)||$, we have
 - $u \cdot v = \frac{1}{4} (||g_*(u) + g_*(v)||^2 - ||g_*(u) - g_*(v)||^2) = \frac{1}{4} (g_*^2(u) + 2g_*(u)g_*(v) + g_*^2(v) - g_*^2(u) + 2g_*(u)g_*(v) - g_*^2(v)) = \frac{4g_*(u)g_*(v)}{4} = g_*(u)g_*(v) = g_*(u) \cdot g_*(v), \forall u, v \in \mathbb{R}^3$

Appendix: SE(3) and preservation of volume

- Why the rigid body motion preserves the volume ?
- Rigid body motion definition:
 - A map $g(t): \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that preserves the norm and the cross-product of any two vectors:
 - $||g_*(v)|| = ||v||$
 - $g_*(u) \times g_*(v) = g_*(u \times v), \forall u, v \in \mathbb{R}^3$
- $g_*(u) \cdot (g_*(v) \times g_*(w)) = u \cdot (v \times w)$
- The triple product corresponds to the volume of the parallelepiped spanned by the three vectors

Appendix: SE(3) and preservation coordinates

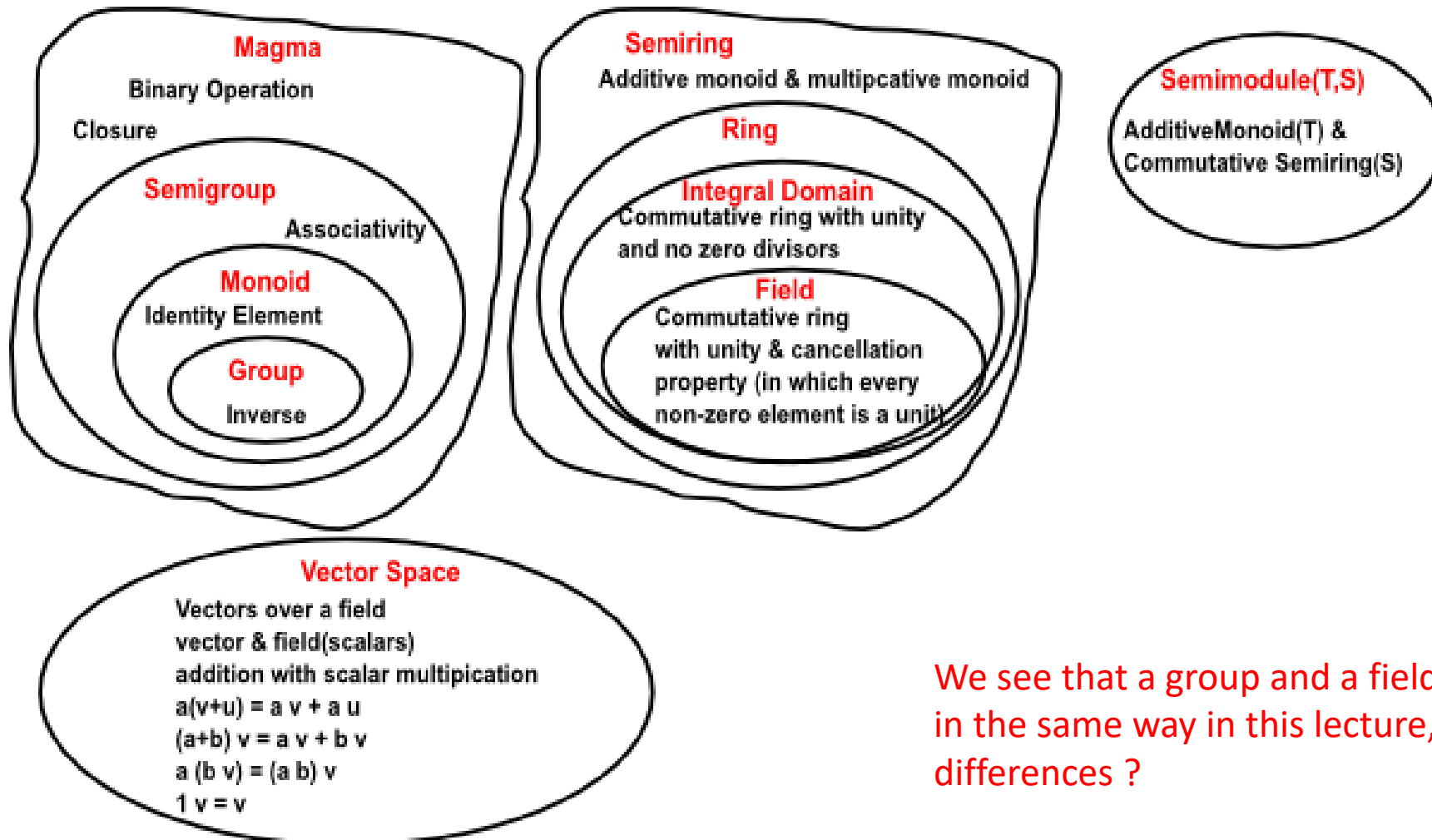
- To show that a rigid body motion preserves the object, it suffices to show that it preserves the coordinates system.
- Let e_1, e_2, e_3 be a basis of $\mathbb{R}^3$
- We have:
 - $g_*(e_i)^T g_*(e_j)^T = \delta_{ij}$
 - $g_*(e_1) \times g_*(e_2) = g_*(e_3)$
- So the three vectors $g_*(e_1), g_*(e_2), g_*(e_3)$ still form a right-handed orthonormal frame.

Appendix: Rigid-body motion $SE(3)$ g vs g_*

- In the slides, g was transformed to g_* when it was applied to vectors.
- This is because, vectors actually do not live in the same space as points, but in the tangent space of a point.
- So it is not exactly the same function, but it behaves in the same way.
- For more info, read Lie Algebras and manifolds, topics too big to be covered in those lectures

Appendix: Other structures

- We didn't define several (less useful for us) structures, in particular rings
- Here is an overview:



We see that a group and a field are not related in the same way in this lecture, can you spot the differences ?